

The Pocket Gophers (Genus *Thomomys*) of Utah

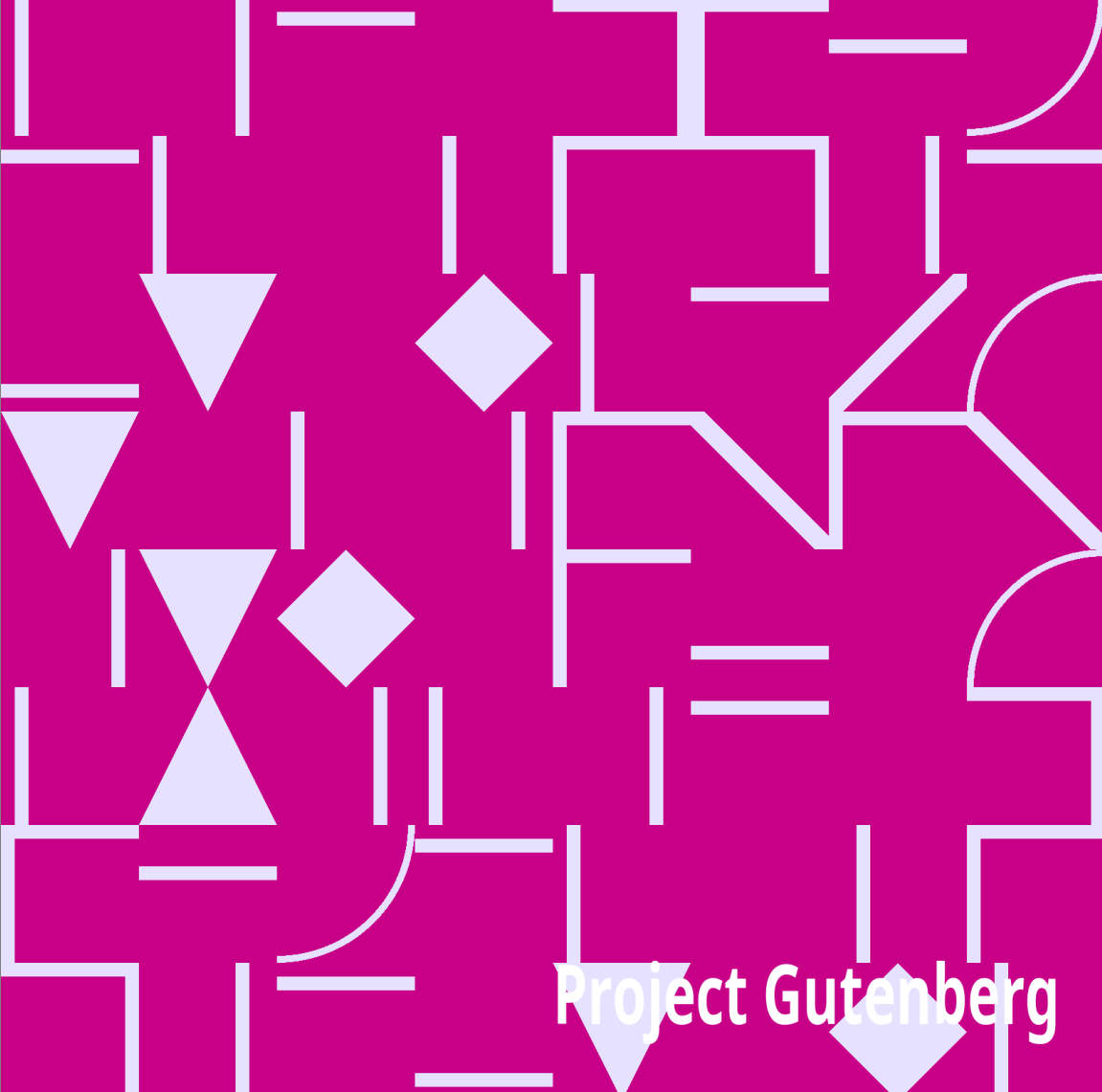
Stephen David Durrant et al.

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*** START OF THE PROJECT GUTENBERG EBOOK THE POCKET GOPHERS (GENUS THOMOMYS) OF UTAH ***

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BY

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**University of Kansas Publications
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**Volume 1, No. 1, pp. 1-82, 1 figure in text
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The Pocket Gophers (Genus *Thomomys*) of Utah

By
STEPHEN D. DURRANT

Contribution from the Department of Biology, University of Utah, and the Museum of Natural History, University of Kansas.

INTRODUCTION

The history of pocket gophers of Utah begins with J. A. Allen's mention in 1874 of mounds of these animals. For them he employed the name "*Thomomys rufescens?*" (1874:65). Actual specimens were reported upon a year later by Elliot Coues (1875:251, 256), who used the name *Thomomys talpoides* for specimens from "Utah" but later in the same paper listed specimens from Provo as *Thomomys talpoides bulbivorus*. Even as the great variation in Utah pocket gophers has been perplexing to modern workers, so it was also to Coues seventy years ago who left the problem with the statement that animals from Provo "exhibit among themselves such variations that their labelling becomes a matter of indifference"! In the same year in another report, Coues and Yarrow (1875:112) used the name *Thomomys talpoides umbrinus* for animals from Provo. In 1877, Coues again referred these same animals to *Thomomys talpoides bulbivorus*, using the name *umbrinus* for the animals of only southern Utah (Coues, 1877:627, 628). The two names *Thomomys bottae* and *Thomomys talpoides*, now applicable to gophers in Utah, were synonymized under the name *Thomomys talpoides bulbivorus* by Coues (1875:256; 1877:627). After this beginning only three other papers, all by J. A. Allen, appeared in the next twenty years. They were reports on collections of mammals made by Walter W. Granger and Charles P. Rowley. One of these contained the description of *Thomomys aureus*. Likewise, in the ensuing twenty years there were only three papers, one in 1901 by C. Hart Merriam in which he described *Thomomys uinta*, one by Allen (1905:119), and Vernon Bailey's (1915) "Revision of the pocket gophers of the genus *Thomomys*" in which he summarized the information then available on these animals within the state. Barnes (1922 and 1927) reprinted the information summarized by Bailey. Since 1927 approximately twenty-five papers, mostly taxonomic, have been published in which reference is made to Utah gophers, and especially since 1930 much information has been accumulated about the distribution and speciation of this genus within the state.

Specimens to the number of 1,045 have been available for this study. Whereas Bailey (*loc. cit.*) listed only four kinds belonging to four different species, thirty-five kinds are now known from Utah. Seven of these are herein described as new. The thirty-five kinds are found to belong to only two instead of four full species.

Inasmuch as the literature is scattered and since names have been applied in different ways at different times, I have attempted to give a synonymy as complete as possible for each form found within the state.

The bibliographies of Hayward (1936 and 1941) and Miller's (1924) "List of North American mammals" have been of great use.

Capitalized color terms in the accounts are after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912.

In the lists of specimens examined, the localities are listed by counties from west to east, beginning at the northwestern corner of the state, and within each county from north to south. When two localities are on the same latitude, the westernmost is listed first.

I am deeply indebted to Professor R. V. Chamberlin, of the University of Utah, for encouragement and support in my investigation. I also acknowledge critical assistance in the preparation of this paper from Professor E. Raymond Hall of the University of Kansas. For the loan of specimens I am grateful to the following: Clinton G. Abbott and Lawrence M. Huey, Natural History Museum of San Diego, San Diego, California; Harold E. Anthony and J. Eric Hill, American Museum of Natural History, New York City, New York; Seth B. Benson, Museum of Vertebrate Zoölogy, University of California, Berkeley, California; William H. Burt, Museum of Zoölogy, University of Michigan, Ann Arbor, Michigan; J. Kenneth Douthett, Carnegie Museum, Pittsburgh, Pennsylvania; Ross Hardy, Dixie Junior College, St. George, Utah; C. Lynn Hayward and Vasco M. Tanner, Brigham Young University, Provo, Utah; H. H. T. Jackson and Viola S. Schantz, United States Fish and Wildlife Service, U. S. National Museum, Washington, D. C.; Remington Kellogg and Alexander Wetmore, U. S. National Museum, Washington, D. C.; J. S. Stanford, Utah State Agricultural College, Logan, Utah.

Unless otherwise indicated, specimens are in the Museum of Zoölogy, University of Utah, Salt Lake City, Utah. In lists of specimens examined, abbreviations are employed as follows:

(A. M. N. H.) American Museum of Natural History.

- (N. H. M. S. D.) Natural History Museum of San Diego.
 (M. V. Z.) Museum of Vertebrate Zoölogy, University of California.
 (U. M.) Museum of Zoölogy, University of Michigan.
 (C. M.) Carnegie Museum.
 (R. H.) Collection of Ross Hardy.
 (B. Y. U.) Brigham Young University.
 (U. S. N. M.) United States National Museum.
 (U. S. A. C.) Utah State Agricultural College.
 (K. U.) Museum of Natural History, University of Kansas.

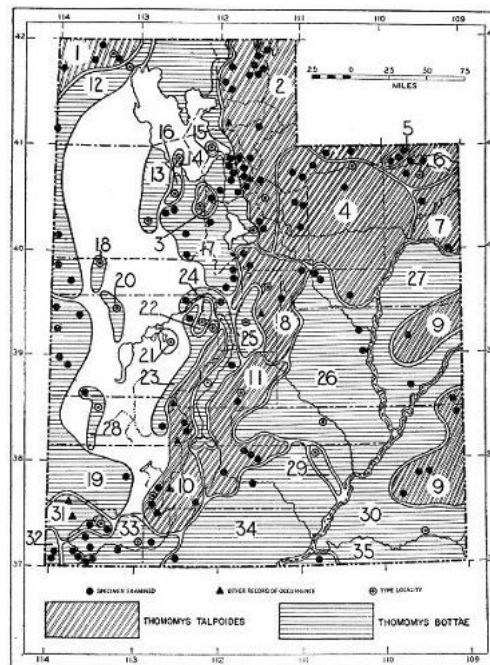


FIG. 1. Map showing the distribution of species and subspecies of pocket gophers in Utah.

- Guide to subspecies:**
- | | | |
|-------------------------------|-------------------------------|-------------------------------|
| 1. <i>T. t. gracilis</i> | 12. <i>T. b. aureiventris</i> | 24. <i>T. b. lenis</i> |
| 2. <i>T. t. wasatchensis</i> | 13. <i>T. b. robustus</i> | 25. <i>T. b. levidensis</i> |
| 3. <i>T. t. oquirrhensis</i> | 14. <i>T. b. minimus</i> | 26. <i>T. b. osgoodi</i> |
| 4. <i>T. t. uinta</i> | 15. <i>T. b. nesophilus</i> | 27. <i>T. b. howelli</i> |
| 5. <i>T. t. pygmaeus</i> | 16. <i>T. b. stansburyi</i> | 28. <i>T. b. wahwahensis</i> |
| 6. <i>T. t. ravus</i> | 17. <i>T. b. albicaudatus</i> | 29. <i>T. b. dissimilis</i> |
| 7. <i>T. t. ocus</i> | 18. <i>T. b. bonnevilliei</i> | 30. <i>T. b. aureus</i> |
| 8. <i>T. t. moorei</i> | 19. <i>T. b. centralis</i> | 31. <i>T. b. birdseyei</i> |
| 9. <i>T. t. fossor</i> | 20. <i>T. b. sevieri</i> | 32. <i>T. b. virgineus</i> |
| 10. <i>T. t. parowanensis</i> | 21. <i>T. b. convexus</i> | 33. <i>T. b. planirostris</i> |
| 11. <i>T. t. levis</i> | 22. <i>T. b. tivius</i> | 34. <i>T. b. absonus</i> |
| | 23. <i>T. b. contractus</i> | 35. <i>T. b. alexandrae</i> |

GENUS **Thomomys** Wied

All pocket gophers of Utah belong to the genus *Thomomys*. There are only two species within the state, *Thomomys bottae* with twenty-four subspecies and *Thomomys talpoides* with eleven subspecies.

Due to marked mutational capacities and ready response to environmental pressures and sedentary habits, pocket gophers differentiate readily into numerous subspecies. It is well known that Utah by its highly varied topography and climate possesses widely different types of habitats. The aforementioned plasticity of these animals and possibly the fact that both species are at the extreme limits of their ranges in Utah account for the numerous forms found within the state.

The genus may be characterized as follows: Highly specialized fossorial rodents, with heavy, thick bodies; all four legs of approximately equal length, but front legs more muscular for digging, and feet provided with long claws; external fur-lined cheek pouches; small eyes, short ears and tail; upper incisors long and projecting external to lips. Skull: Stout and flattened; zygomatic arches well developed and usually widely spreading; all teeth with permanent pulp cavities; incisors superficially smooth, but fine median groove present on anterior face of each upper incisor; dental formula, i. 1/1, c. 0/0, p. 1/1, m. 3/3; external auditory canal long; stapedial artery small and enclosed within an osseous canal.

Thomomys talpoides (Richardson)

Thomomys talpoides is a northern species that in Utah approaches the southern limits of its range. The animals of this species inhabit the mountains and high valleys. In the southward extension of their range, as in Utah, they are found at higher elevations which zonally represent lower elevations at more northern latitudes. The specific characters are: Sphenorbital fissure absent; incisive foramina anterior to infraorbital canal; anterior prism of P4 triangular; interparietal relatively large; lambdoidal suture concave posteriorly in region of interparietal, in Utah specimens.

Thomomys talpoides gracilis Durrant

Thomomys quadratus gracilis Durrant, Bull. Univ. Utah, 39 (No. 6):3, February 28, 1939.

Thomomys talpoides gracilis Durrant, Bull. Univ. Utah, 30 (No. 5):6, August 24, 1939; Goldman, Journ. Mamm., 25:414, December 12, 1944.

Thomomys quadratus fisheri Hall, Univ. California Publ. Zoöl., 37:4, April 10, 1931.

Thomomys uinta Bailey, N. Amer. Fauna, 39:114, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):83, April, 1922; Bull. Univ. Utah, 17 (No. 12):104, June, 1927.

Type.—Male adult, skin and skull; No. 44866, Museum of Vertebrate Zoölogy, University of California; Pine Canyon, 6,600 ft., 17 mi. NW Kelton, Box Elder County, Utah; July 12, 1930; collected by Annie M. Alexander; original number 676.

Range.—Mountainous regions of extreme northwestern Utah.

Diagnosis.—Size medium (see measurements). Color: Upper parts Buckthorn Brown grading over the sides and flanks to Light Buff on the underparts; chin white; nose and postauricular patches grayish black. Claws on front feet long and slender. Skull: Long and slender; rostrum long and narrow; zygomatic and mastoidal breadths slight; palatal pits deep; upper incisors narrow; basioccipital wide.

Comparisons.—Compared with topotypes of *Thomomys talpoides fisheri*, *gracilis* is of approximately the same size. Upper parts darker and underparts lighter; postauricular patches larger and darker; claws on front feet longer and slenderer. Skull: Generally longer and narrower; nasals and rostrum longer; basioccipital wider.

As compared with *T. t. uinta*, *gracilis* is of approximately the same size but differs as follows: Color: Lighter throughout; postauricular patches markedly smaller and lighter; inguinal and pectoral regions much lighter. One characteristic difference is in the ear. In *uinta* the external opening of the ear is much larger; the pinna of the ear is larger, more rounded at the tip, and lacks most of the pigmentation on the inner margin. Skull: Generally narrower and longer; nasals longer; zygomatic arches weaker and less angular; upper incisors narrower.

This form is easily distinguished from *bridgeri* by smaller size, and by the skull being longer, narrower and less angular.

From *Thomomys talpoides oquirrhensis* to the southeast, *T. t. gracilis* can be distinguished by: Total length and ear shorter. Color: Generally lighter, except the underparts which are about the same; postauricular patches larger and more deeply pigmented. Skull: Braincase less inflated; nasals truncated posteriorly as opposed to rounded; zygomatic and mastoidal breadths less; rostrum shorter but narrower; upper incisors narrower and shorter.

For comparisons with *wasatchensis* see comparisons under that form.

In general, this mountain form can be distinguished from all other *talpoides* in Utah by lighter color, narrow, slender, "graceful" skull whence the name *gracilis* is derived.

Remarks.—In Utah, *gracilis* is limited to the extreme northwestern corner of the state. This part of the state is in the Snake River drainage. The main part of the range of this race lies in south-central and southwestern Idaho and northeastern Nevada. The center of its range might be considered to be in the Jarbidge Mountains area of Nevada. The south slopes of these mountains are in the Humboldt River drainage, while the north slopes are in the Snake River drainage, and this subspecies occurs as far north as the Snake River and south and west almost to central Nevada. No specimens are available from the area in Utah between the Raft River Mountains inhabited by *gracilis* and the Wasatch Mountains in central Utah inhabited by *wasatchensis*. Judging from the nature of the terrain, the range of *gracilis* does not extend eastward much beyond the Raft River Mountains. The type locality for a gopher of a different species, *Thomomys bottae aureiventris*, is in the first valley east of these mountains. Furthermore, all valleys to the east and south, as far as known, are inhabited by gophers of the *bottae* group. Also, all mountain ranges in this area, as far east as the Wasatch Mountains are inhabited by members of the *bottae* group.

No specimens from Utah indicate intergradation between *gracilis* and *wasatchensis*, the form to the east, but specimens from farther north at Albion, Cassia County, Idaho, do show intergradation. Bailey (1915:116), Hall (1931:4), and Durrant (1939:6) have reported on these specimens which at the present time seem best referred to *T. t. gracilis*.

Specimens examined.—Total, 24, distributed as follows: Box Elder County: Yost, 4 (U. S. A. C.); Pine Canyon, 6,600 ft., 17 mi. NW Kelton, 7 (M. V. Z.); Lynn Canyon, Raft River, 4; Park Valley, 3 (U. S. A. C.); Etna, 4 (U. S. A. C.); Raft River Mountains, Clear Creek Camp of Minnedoka National Forest, 1 (R. H.); Raft River Mountains, 1,500 feet above Clear Creek Camp of Minnedoka National Forest, 1 (R. H.).

***Thomomys talpoides wasatchensis* new subspecies**

Thomomys quadratus uinta Hall, Univ. California Publ. Zoöl., 37:4, April 10, 1931.

Thomomys talpoides uinta Goldman, Journ. Mamm., 20:234. May 14, 1939.

Thomomys uinta Bailey, N. Amer. Fauna, 39:114, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):83, April, 1922; Bull. Univ. Utah, 17 (No. 12):104, June, 1927; Stanford, Journ. Mamm., 12:360, November 11, 1931.

Type.—Male, adult, skin and skull, No. 1604, Museum of Zoölogy, University of Utah; Midway, 5,500 ft., Wasatch County, Utah; September 1, 1936; collected by S. D. Durrant; original number 1049.

Range.—Wasatch Mountains and neighboring high valleys as far south as Spanish Fork Canyon, Utah County.

Diagnosis.—Size medium (see measurements). Color: Upper parts Snuff Brown, finely mixed with black; sides and flanks Sayal Brown; underparts overlaid with Cinnamon Buff, with suffusion of black on underfur; postauricular patches black, extending around ear; ears pointed and covered with black hairs; nose, cheeks, chin and top of head dusky; front feet, hind feet and distal part of tail white; tail covered proximally with light brown hairs. Skull: Moderately heavy and ridged; nasals long, wide posteriorly and not markedly dilated distally;

posterior ends of nasals emarginate; zygomatic arches fairly widely spreading and angular, being nearly straight in adults, but tending to bow out slightly at posterior ends in young; zygomatic processes of maxillae heavy; interparietal small and variously shaped, but always wider than long; interorbital region fairly wide; well marked dorsal depression in frontals posterior to ends of nasals; interpterygoid space narrowly V-shaped; tympanic bullae large; occipital condyles large and widely separated; foramen magnum large and higher than wide; basioccipital wide; dentition light.

Comparisons.—From topotypes of *Thomomys talpoides moorei*, *wasatchensis* differs as follows: Size slightly larger; ears longer and more pointed. Color: Generally darker throughout; postauricular patches smaller. Skull: Zygomatic arches not as widely spreading; zygomatic processes of squamosals dip farther ventrally; premaxillae less extended posterior to nasals; nasals wider posteriorly and less dilated distally; median dorsal depression of frontals present; tympanic bullae generally larger, but less inflated ventrally; foramen magnum larger especially in dorsoventral dimension; occipital condyles farther apart; basioccipital wider; alveolar length of upper molar series less; molariform teeth smaller; upper incisors wider and shorter.

Topotypes of *wasatchensis* differ from topotypes and near topotypes of *Thomomys talpoides uinta* as follows: Size larger in every measurement taken. Color: Darker throughout; ears longer and more pigmented; opening of external ear smaller; postauricular patches larger. Skull: In females larger throughout, more massive and angular; nasals longer, wider and not so dilated distally; rostrum longer but wider; zygomatic arches wider, more angular and less widely spreading posteriorly; extension of premaxillae posterior to nasals less; tympanic bullae larger, but less inflated ventrally; foramen magnum larger and more ovoid; width across occipital condyles greater; basioccipital wider; molariform teeth smaller; upper incisors shorter and wider.

Topotypes of *wasatchensis* can be distinguished from those of *Thomomys talpoides oquirrhensis* as follows: Size larger; tail longer; ears longer. Color: Slightly darker on sides and underparts. Skull: Heavier, more ridged and angular; nasals more dilated distally; posterior ends of nasals more deeply emarginate; zygomatic arches heavier and more widely spreading, but more nearly parallel and less divergent posteriorly; zygomatic processes of maxillae much heavier; braincase and tympanic bullae larger; pterygoid hamulae shorter; interpterygoid space more narrowly V-shaped; wider across occipital condyles; foramen magnum larger and more ovoid.

From topotypes of *Thomomys talpoides gracilis*, *wasatchensis* differs as follows: Size larger; hind foot longer; ears longer and more pointed. Color: Darker throughout; postauricular patches relatively smaller. Skull: Larger, heavier and more angular; nasals emarginate posteriorly as opposed to truncate; rostrum heavier; zygomatic arches heavier and more widely spreading; zygomatic processes of maxillae much heavier and more angular; mastoid breadth greater; interparietal relatively smaller; extension of premaxillae posterior to nasals actually as well as relatively less; palatal pits deeper; tympanic bullae larger; interpterygoid space more narrowly V-shaped; foramen magnum more ovoid; upper incisors wider.

Topotypes of *wasatchensis* can be readily distinguished from those of *Thomomys talpoides levis* and *parowanensis* by larger size; more massive, ridged, angular skulls; larger tympanic bullae; large, ovoid foramen magnum; and relatively smaller interparietal.

Remarks.—Specimens from Mount Timpanogos and environs are intergrades between *moorei* and *wasatchensis*. They resemble *moorei* in the shape and size of the tympanic bullae, and are intermediate in the size and shape of the foramen magnum. In the majority of characters they resemble *wasatchensis* to which they are here referred. The animals from east of Salt Lake City in Salt Lake County are intergrades between *oquirrhensis* and *wasatchensis* and show some characters of *uinta*, but are referable to *wasatchensis*. Animals from Morgan County and western Summit County are intergrades between *wasatchensis* and *uinta*. They resemble *uinta* in size, shape of nasals and size of tympanic bullae. The remainder of the cranial details place them with *wasatchensis*. Morphologically the animals from Wellsville, Cache County, were the closest to the topotypes of any obtained and are nearly indistinguishable from them. Like the topotypes of *wasatchensis* this population inhabits a high valley. The remaining specimens from Cache County resemble those from Morgan and Summit counties.

Specimens examined.—Total, 119, distributed as follows: *Cache County*: Logan Canyon, Beaver Basin, Utah-Idaho Line, 2 (U. S. A. C); Logan Canyon, Tony Grove Camp, 6 (U. S. A. C); Logan Canyon, Green Camp, 3 (U. S. A. C); Logan Canyon, 3 (U. S. A. C); Logan Mountains, 20 mi. E Logan, 3 (U. S. A. C); Logan Peak area, 13 (U. S. A. C); near Providence Peak, Logan Mountains, 1 (U. S. A. C.); Wellsville, 10 (U. S. A. C); Hardware Ranch, Blacksmith Fork, 1 (U. S. A. C); Avon, 1 (U. S. A. C); 1 mi. E Avon, 1 (U. S. A. C); 7-8 mi. E Avon, 1 (U.

S. A. C.). *Weber County*: South Fork, Ogden River, 18 mi. E Ogden, 4 (M. V. Z.). *Morgan County*: East Canyon, 18 mi. NW Park City, 6,000 ft., 1. *Davis County*: 8 mi. NE Salt Lake City, 1. *Salt Lake County*: Mouth of Dry Canyon, 1 mi. NE Salt Lake City, 1; 4 mi. above mouth City Creek Canyon, 5,000 ft., 1; mouth of Emigration Canyon, 1; mouth of Millcreek Canyon, 1; Lambs Canyon, 13 mi. SE Salt Lake City, 2 (C. M.); mouth of Big Cottonwood Canyon, 1. *Summit County*: Park City, 1 (U. S. N. M.). *Wasatch County*: Midway, 5,500 ft., 29. *Utah County*: Mt. Timpanogos, 1 mi. N Aspen Grove, 7,500 ft., 20; Aspen Grove, Mt. Timpanogos, 5 (1, U. S. A. C.; 4, B. Y. U.); Head of Grove Creek, Mt. Timpanogos, 4 (B. Y. U.).

Additional Records: *Weber County*: Ogden, 6. *Salt Lake County*: Parleys Canyon, 1 (Bailey, 1915:114).

***Thomomys talpoides oquirrhensis* Durrant**

Thomomys talpoides oquirrhensis Durrant, Bull. Univ. Utah, 30 (No. 5):3, October 24, 1939.

Type.—Male, adult, skin and skull; No. 2605, Museum of Zoölogy, University of Utah; Settlement Creek, Oquirrh Mountains, 6,500 ft., Tooele County, Utah; June 11, 1938; collected by S. D. Durrant; original number 1461.

Range.—Known only from the Oquirrh Mountains, which are in Salt Lake, Tooele and Utah counties, Utah.

Diagnosis.—Size medium (see measurements); ear long; tail short, claws of front feet long and slender. Color: Upper parts Buckthorn Brown, mixed with black, grading over the sides and flanks to Pinkish Buff on the ventral surface; feet white; nose grayish black; postauricular patches medium in size and black; chin and throat with varying amounts of white; proximal two-thirds of tail dark brown, distal third white. Skull: Long and slender, but relatively wide across mastoidal region; nasals long and rounded posteriorly; rostrum long and narrow; zygomatic arches weak and not widely spreading, tending to be slightly bowed out posteriorly, but in the main roughly parallel to the sides of the skull; outer margin of zygomatic arch slightly concave, and zygomatic arch dips deeply ventrad; dorsal surface of skull smooth, with weakly defined parietal crests; parietal crest nearly parallel, but bowed medially, in parietal region, and flaring widely posteriorly to pass lateral to interparietal; tympanic bullae large, truncate anteriorly and markedly inflated ventrally; upper incisors short and fairly robust.

Comparisons.—From *Thomomys talpoides uinta*, *oquirrhensis* may be differentiated as follows: Color: Darker throughout; postauricular patches larger and darker; ears longer and more pointed; inner margin of pinna heavily pigmented; external opening of ear smaller. Skull: Nasals rounded posteriorly rather than deeply emarginate, and less flaring distally; zygomatic arches weaker and markedly less widely spreading; pterygoid hamulae weaker; basisphenoid narrower; upper incisors shorter and wider.

For comparisons between *oquirrhensis* and *Thomomys talpoides gracilis*, and *oquirrhensis* and *wasatchensis*, see comparisons under those forms.

Topotypical specimens of *oquirrhensis* can be distinguished from those of *Thomomys talpoides moorei* as follows: Color generally darker, due to greater admixture of black; terminal bands of hair actually lighter; postauricular patches larger and darker; ears longer, more pointed and with more heavily pigmented pinnae; tail shorter. Skull: About the same size; smoother; zygomatic arches weaker and less widely spreading; nasals rounded posteriorly as opposed to emarginate; mastoid breadth less; pterygoid hamulae weaker; upper incisors wider.

Remarks.—This race is limited to the Oquirrh Mountains, a high mountain range that lies parallel to, and just west of the Wasatch Mountains, in Utah, Salt Lake and Tooele counties. These mountains were connected in past times to the Wasatch Mountains by the Transverse Range, and by a sand and gravel bar deposited by Pleistocene Lake Bonneville. The Jordan River in its course from Utah Lake to the Great Salt Lake has cut a channel through the aforementioned bar. This channel has been cut to the level of the surrounding valleys as is indicated by the meandering nature of the stream through this part of its course. As a result the Oquirrh Mountains are relatively isolated. Although separated from the Wasatch Mountains by the Jordan River Valley only a few miles wide, the pocket gophers are distinct on each mountain. A population of *T. bottae* is interposed between the two mountain ranges as is indicated by specimens from Riverton, six miles north of the Transverse Range. The populations of *bottae* are subspecifically the same on the two sides of the Jordan River.

On the east side of the Oquirrh Mountains, pocket gophers collected from the Jordan Valley up Rose Canyon to about 5,000 feet elevation were all of the species *T. bottae*. Between 5,000 and 6,000 feet there is an area in which the ranges of *bottae* and *talpoides* overlap. When trapping, it is possible to predict what species will be taken by

the types of burrows and soil. Gophers of the *bottae* group have their burrows in the areas of the deepest soil and heaviest vegetation, whereas the areas of shallow, rocky soil covered with sparse vegetation are the habitat of *talpoides*. Above 6,000 feet the only gopher encountered is *talpoides*. Along Settlement Creek on the west side of the Oquirrh Mountains, which is the type locality of *oquirrhensis*, *bottae* and *talpoides* have essentially the same vertical distribution as in Rose Canyon. On this mountain the two species appear to be in competition.

The available information, based on collections, indicates that the Oquirrh Mountains are the only mountains west of the Wasatch Range upon which *talpoides* occurs. In Utah, all other mountains to the west, as far as known, are inhabited by subspecies of *Thomomys bottae*.

Specimens examined.—Total, 41, as follows: *Tooele County*: Settlement Creek, Oquirrh Mountains, 6,500 ft., 14. *Salt Lake County*: Rose Canyon, Oquirrh Mountains, 5,650 ft., 27.

***Thomomys talpoides uinta* Merriam**

Thomomys uinta Merriam, Proc. Biol. Soc. Washington, 14:112, July 19, 1901; Bailey, N. Amer. Fauna, 39:113, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):83, April, 1922; Bull. Univ. Utah, 17 (No. 12):104, June, 1927; Stanford, Journ. Mamm., 12:360; November 11, 1931; Goldman, Journ. Washington Acad. Sci., 28:333, July 15, 1938; Davis, The Recent mammals of Idaho, pp. 239, 259, The Caxton Printers, Ltd., Caldwell, Idaho, April 5, 1939.

Thomomys talpoides uinta Goldman, Journ. Mamm., 20:234, May 14, 1939.

Thomomys quadratus uinta Hall, Univ. California Publ. Zool., 37:4, April 10, 1931.

Type.—Male, adult, skin and skull, No. 22501/30051, U. S. National Museum (Biological Surveys Collection); north base Gilbert Peak, Uinta Mountains, 10,000 ft., Summit County, Utah; June 6, 1890; collected by Vernon Bailey; original number 1262 (after Merriam, type not seen).

Range.—Uinta Mountains in Duchesne County, eastern Wasatch and Summit counties, and western Uintah County south to the Roan, Brown and Book cliffs in Carbon County.

Diagnosis.—Size medium (see measurements). Color: Upper parts Snuff Brown finely mixed with black, paling over sides and flanks to near Pinkish Buff on underparts; postauricular patches relatively small and dusky; external opening of ear large; pinnae usually lightly pigmented; hind feet white; front feet usually white only at base of toes; distal third to half of tail white; tail usually light below, with proximal dorsal half covered with darker hairs; nose, chin, cheeks and top of head dusky; usually considerable white on throat. Skull: Small, slender, and not heavily ridged; nasals short and dilated distally; posterior margins of nasals emarginate; zygomatic arches moderately widely spreading, widest posteriorly; interparietal pentagonal or subquadrangular; interpterygoid space V-shaped; tympanic bullae well inflated ventrally; upper incisors long and narrow.

Comparisons.—For comparisons with other subspecies of *Thomomys talpoides*, see accounts of those forms.

Remarks.—The range formerly ascribed to *uinta* (Bailey, 1915:114; Barnes, 1922:83, 1927:104) is now known to be inhabited by animals belonging to three distinct subspecies. The range of *uinta* as now understood is restricted to the southern and western parts of the Uinta Mountains and their environs. Three specimens from the Book Cliffs, Sunnyside, Carbon County, are not typical, but in a majority of their characters agree with *uinta* to which they are here referred.

I have seen only one specimen from the type locality. It is one of the series on which Merriam (1901:112) based his original description. In addition, I have studied several large series of near topotypes. From the material at hand, and from Merriam's description (*loc. cit.*), I regard the animals on which the name *uinta* was based as intergrades between *Thomomys talpoides ravus*, the race to the northeast, on the one hand and the animals of the western and southern parts of the Uinta Mountains on the other hand. The affinities of the type series are with the animals from the latter area which are here all referred to *uinta*.

Specimens examined.—Total, 41, distributed as follows: *Summit County*: 2 mi. S junction Bear River and Haydens Fork, 2 (C. M.); N base, Gilbert Peak, 10,000 ft., 1 (U. S. N. M.); Smith and Moorehouse Creek, 2; Bald Peak, 25 mi. NE Kamas, 15 (8, M. V. Z.; 6, C. M.). *Duchesne County*: Petty Mountain, 15 mi. N Mountain Home, 9,500 ft.,

6 (C. M.). *Wasatch County*: Wolf Creek Pass, 18 mi. NW Hanna, 1 (U. S. A. C.); Lost Lake, Uinta Mountains, 10 (B. Y. U.); Current Creek, Uinta Mountains, 1 (U. S. N. M.). *Carbon County*: Forks, Sunnyside, 9,000 ft., 3.

Additional records.—*Summit County*: Uinta Mountains, 6 (see Bailey, 1915:114).

***Thomomys talpoides pygmaeus* Merriam**

Thomomys pygmaeus Merriam, Proc. Biol. Soc. Washington, 14:115. July 19, 1901.

Thomomys talpoides pygmaeus Davis, The Recent mammals of Idaho, p. 252, The Caxton Printers, Ltd., Caldwell, Idaho, April 5, 1939.

Type.—Male, adult, skin and skull, No. 55251, U. S. National Museum (Biological Surveys Collection); 10 mi. NE Montpelier, in open sagebrush of Transition Zone, 6,600 ft., Bear County, Idaho; July 29, 1893; collected by Vernon Bailey; original number 4150 (after Merriam, type not seen: see, also, Bailey, 1915:109).

Range.—Limited to Daggett County.

Diagnosis.—Size: Small (see measurements). Color: Upper parts near Bister slightly mixed with black, grading over sides and flanks to Ochraceous Buff on underparts; postauricular patches small and dusky; hind feet white; front feet dusky, being white only at base of claws; chin and nose dusky; tail brown, lighter below and tipped with white. Skull: Very small, slender and smooth; nasals short and slender; zygomatic arches weak and not widely spreading; rostrum narrow; extension of premaxillae posterior to nasals short; parietal ridges hardly noticeable; interparietal large; extension of supraoccipital posterior to lambdoidal suture long; tympanic bullae actually small, but relatively large; basioccipital narrow; interpterygoid space narrow and acutely angled; upper incisors markedly recurved; molariform teeth relatively large.

Comparisons.—This small pocket gopher can be distinguished from all other members of *Thomomys talpoides* occurring in Utah by remarkably small size, and slender, weak, small skull with strongly recurved upper incisors.

Remarks.—The specimens used in this study were those recorded by Svihla (1931:261). She reports that they were obtained in the flood-plain banks of the streamsides, and preferred the pine belt. This shows probably an extension of range with reference to life zones, as heretofore the main reported localities of capture have been in sagebrush in the Transition Life-zone.

Insofar as I am aware, Mrs. Svihla's specimens are the only ones of this subspecies ever obtained in Utah. Additional work is necessary in southwestern Wyoming to outline accurately the geographic distribution of this subspecies. In comparison with topotypes, the specimens from Utah are lighter in color and some specimens have slightly larger skulls, suggesting slight intergradation with *Thomomys talpoides uinta*.

Specimens examined.—Total, 18 (all in Museum of Zoölogy, University of Michigan), distributed as follows: *Daggett County*: Sheep Creek, 4; 1 mi. W Summit Springs, 4; Beaver Creek, 22 mi. S Manila, 9; Granite Park, 24 mi. S Manila, 1.

***Thomomys talpoides ravis* new subspecies**

Type.—Male, adult, skin and skull, No. 13690, Carnegie Museum; Vernal-Manila Highway, 19 mi. N Vernal, 8,000 ft., Uintah County, Utah; August 22, 1937; collected by J. K. and M. T. Doult; original number 4718.

Range.—Uinta Mountains in Daggett, northern Uintah and northern Summit counties.

Diagnosis.—Size large (see measurements); ears relatively narrow; hind foot relatively small. Color: Upper parts between Drab and Light Drab, darkest along middorsal line due to mixture of hairs tipped with light brown; sides and flanks Light Drab; entire underparts creamy white; front and hind feet, ventral surface of tail and end of tail white; proximal two-thirds of tail covered dorsally with light brown hairs; nose and cheeks dusky; postauricular patches black. Skull: Large, heavy and ridged; rostrum long and narrow; nasals long, moderately dilated distally and with a distal hump; posterior ends of nasals emarginate; parietal and lambdoidal crests well developed; zygomatic arches moderately heavy and widely spreading, widest posteriorly; zygomatic processes of maxillae moderately heavy and flaring abruptly from base of rostrum; marked middorsal depression in frontals present;

interparietal pentagonal; extension of premaxillae posterior to nasals long; posterior tongues of premaxillae long, slender and rounded proximally; braincase high, vaulted and relatively narrow; tympanic bullae well inflated ventrally, and ridged in old animals; pterygoid hamulae long; interpterygoid space narrowly V-shaped; upper incisors long and narrow; molariform teeth medium.

Comparisons.—Compared with topotypes of *Thomomys talpoides bridgeri*, *ravus* differs as follows: Size larger; hind foot smaller; ears narrower. Color: Lighter throughout, grayish as opposed to brown. Skull: Smaller, narrower, less angular and less massive; nasals, rostrum, zygomatic processes of maxillae, ascending branches of premaxillae and posterior tongues of premaxillae all narrower; extension of premaxillae posterior to nasals longer; interparietal wider; braincase higher and narrower; tympanic bullae approximately the same size, but more inflated ventrally; interpterygoid space more narrowly V-shaped; upper incisors narrower; molariform teeth weaker.

Compared with topotypes and near topotypes of *Thomomys talpoides uinta*, *ravus* differs as follows: Size larger in every measurement taken. Color: Lighter throughout, being grayish as opposed to brown. Skull: Larger in every measurement taken; rostrum and nasals actually as well as relatively longer; extension of premaxillae posterior to nasals longer; upper incisors longer and wider; molariform teeth larger.

There is only one other gray subspecies of *Thomomys talpoides* in Utah, *Thomomys talpoides ocus*. Topotypes of *ravus* differ from it as follows: Size markedly larger in every measurement taken. Color: Darker, more brown hairs. Skull: Larger in every measurement taken; premaxillae extended farther posteriorly to nasals; extension of supraoccipital posterior to lambdoidal suture markedly less; tympanic bullae actually as well as relatively smaller; upper incisors longer and more procumbent.

This new subspecies can be readily distinguished from all other subspecies of *Thomomys talpoides* occurring in Utah by markedly greater size and paler, more grayish color.

Remarks.—The range of this form appears to be limited to the north slopes of the Uinta Mountains, except in Daggett County where it occurs also on the south slopes. Intergradation in color and in cranial details with *bridgeri* is shown by animals from the East Fork of Blacks Fork, thirty-one miles SSW Fort Bridger, and by those from Henrys Fork, 8,300 ft., both in Summit County. Due to the grayish color and the narrower, weaker skull they are referred to *ravus*. Intergradation with *uinta* is shown by specimens from the type locality of the latter race. The type series of *uinta* consists of intergrades between *ravus* and the animals to the west and south (see remarks under *uinta*).

It is doubtful whether *bridgeri* occurs in Utah. Material from Rich County and extreme northern Cache County would settle the question. Perhaps *bridgeri* is restricted to the lower valleys in southwestern Wyoming. Two specimens from northern Cache County, from Logan Canyon, Beaver Basin, Utah-Idaho Line appear to be intergrades between *bridgeri* and *wasatchensis*, but are referable to the latter race.

Specimens examined.—Total, 38, distributed as follows: *Summit County*: Henrys Fork, 8,300 ft., 8; E Fork, Blacks Fork, 31 mi. SSW Fort Bridger, 4 (C. M.). *Daggett County*: Vernal-Manila Road, 4 mi. W Green's Lake, 7,500 ft., 6 (C. M.); Elk Park, Uinta Mountains, 5 (B. Y. U.). *Uintah County*: Trout Creek, SE Trout Peak, 22 mi. NW Vernal, 9,300 ft., 5 (C. M.); Vernal-Manila Highway, 19 mi. N Vernal, 8,000 ft., 6 (C. M.); Taylor Peak, 17 mi. N Vernal, 4 (C. M.).

***Thomomys talpoides ocus* Merriam**

Thomomys clusius ocus Merriam, Proc. Biol. Soc. Washington, 14:114, July 19, 1901.

Thomomys clusius Allen, Bull. Amer. Mus. Nat. Hist., 13:246, November 25, 1896.

Thomomys ocus Bailey, N. Amer. Fauna, 39:107, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):83, April, 1922; Bull. Univ. Utah, 17 (No. 12):102, June, 1927.

Type.—Male, adult, skin and skull, No. 18852/25586, U. S. National Museum (Biological Surveys Collection); dry sagebrush mesas at Harveys Ranch, Smiths Fork, 6 mi. SW Fort Bridger, 6,657 ft., Uinta County, Wyoming; May 24, 1890; collected by Vernon Bailey; original number 1194 (after Bailey, type not seen).

Diagnosis.—Size small (see measurements). Color: Upper parts Tilleul Buff overlaid with Avellaneous, grading over sides and flanks to nearly white on underparts; underparts with faint wash of creamy white; postauricular

patches small and dusky and completely circling the ear; nose and cheeks dusky; front feet, hind feet, throat, ventral surface of tail and distal half of tail white. Skull: Small, slender but compact; nasals rounded posteriorly; extension of premaxillae posterior to nasals very short; zygomatic arches robust, but not widely spreading, widest posteriorly; interparietal large and pentagonal in shape; extension of supraoccipital posterior to lambdoidal suture long; tympanic bullae actually as well as relatively large; basioccipital narrow; pterygoid hamulae long and ridged; upper incisors short and strongly recurved.

Comparisons.—Compared with one topotype and seven near topotypes of *Thomomys talpoides pygmaeus*, *ocius* differs as follows: Size larger in every measurement taken. Color: Lighter throughout, grayish as opposed to brown; distal half of tail white as opposed to only a few white hairs at tip of tail. Skull: Larger in every measurement taken; skull more compact; zygomatic arches heavier and more widely spreading posteriorly; tympanic bullae larger; upper incisors larger, but equally strongly recurved; molariform teeth larger.

Topotypes of *ocius* can be distinguished from those of *Thomomys talpoides uinta* as follows: Color: Lighter throughout, grayish as opposed to brown. Skull: Nasals rounded posteriorly as opposed to emarginate; zygomatic arches more robust; interparietal pentagonal as opposed to subquadrangular; extension of supraoccipital posterior to lambdoidal suture markedly greater; tympanic bullae actually as well as relatively much larger; upper incisors short and strongly recurved as opposed to long and procumbent.

Specimens of this subspecies can be distinguished from all other members of the species *Thomomys talpoides* occurring in Utah by their grayish color, and by small, compact skulls with very large tympanic bullae and short strongly recurved upper incisors.

Remarks.—Two specimens from Vernal, Uintah County, are intergrades between *ocius* and *uinta*. They resemble *uinta* in size and dorsal color, but are slightly lighter tending toward the color of *ocius*. Ventrally they are intermediate in color but more like *ocius*. The skulls are more like those of *ocius* in general appearance, extension of supraoccipital posterior to the lambdoidal suture, shape and thickness of the zygomatic arches, posterior tongues of premaxillae, size of tympanic bullae and recurved upper incisors. They more closely resemble *uinta* in shape of posterior ends of nasals, basioccipital and shape of the zygomatic processes of the squamosals. In all of the above mentioned characters, they are intermediate between the two named forms, but tend towards one or the other as listed. The majority of characters are more as in *ocius* to which they are here referred.

When Goldman (1939:233, 234) listed the named subspecies of *Thomomys talpoides*, he hesitated to include *ocius* and merely mentioned that *ocius*, *pygmaeus* and *idahoensis* might also belong to *talpoides*. Davis (1939:240, 241) found intergradation between *idahoensis* and *fuscus* and also between *idahoensis* and *pygmaeus*, and, therefore, arranged the last two mentioned forms as subspecies of *talpoides*. This present study reveals intergradation between *ocius* and *uinta*, and also between *ocius* and *fossor* (see account of *fossor*). Therefore, *ocius* is properly to be treated as a subspecies of the series of intergrading forms of which *talpoides* is the earliest named.

All specimens of *ocius* known from Utah are from the extreme eastern part of the northeastern corner of the state. The type locality of *ocius* is near Fort Bridger, Wyoming, which is north of Utah. I have seen one specimen from 12 miles west of Linwood, Daggett County, Utah, on Henrys Fork in Wyoming. Additional collecting in northern Utah probably will reveal *ocius* to inhabit also parts of northern Utah.

Specimens examined.—Total, 4, distributed as follows: *Uintah County*: Vernal, 2 (C. M.); Uncompahgre Indian Reservation, 2 (A. M. N. H.).

***Thomomys talpoides moorei* Goldman**

Thomomys fossor moorei Goldman, Journ. Washington Acad. Sci., 28:335, July 15, 1938.

Thomomys talpoides moorei Goldman, Journ. Mamm., 20:234, May 14, 1939.

Type.—Male, adult, skin and skull, No. 248222, U. S. National Museum (Biological Surveys Collection); 1 mi. S Fairview, 6,000 ft., Sanpete County, Utah; February 19, 1928; collected by A. W. Moore; X-catalogue number 24799 (after Goldman, type not seen).

Range.—Wasatch Plateau in Sanpete, Utah, Carbon and Emery counties, and in Wasatch Mountains south of Spanish Fork Canyon.

Diagnosis.—Size medium (see measurements). Color: Upper parts between Cinnamon and Sayal Brown, with mixture of black hairs, grading through Cinnamon on sides and flanks to Pale Pinkish Buff on underparts, clearest on inguinal and pectoral regions; nose and cheeks dusky; postauricular patches medium in size and black; ears black; chin buffy white; front and hind feet white; tail mostly white with brownish hairs on dorsal surface. Skull: Large, robust; nasals long and deeply emarginate on posterior ends, and dilated distally; zygomatic arches robust and widely spreading; zygomatic processes of maxillae heavy; interparietal comparatively small, but always wider than long; extension of premaxillae posterior to nasals short; tympanic bullae moderate in size, but markedly inflated ventrally; pterygoid hamulae long; interpterygoid space narrowly V-shaped; upper incisors long and moderately recurved; molariform teeth light.

Comparisons.—Topotypes of *moorei* differ from topotypes and near topotypes of *Thomomys talpoides uinta* as follows: Size slightly larger. Color: Upper parts and sides lighter; tail lighter; postauricular patches larger and darker; ears more pointed, smaller and darker. Skull: Larger, heavier and more massive; nasals longer, but deeply emarginate posteriorly as in *uinta*; rostrum wider and longer; zygomatic arches heavier and more angular; zygomatic processes of maxillae heavier; interparietal generally smaller and shorter; braincase wider; tympanic bullae more inflated ventrally; interpterygoid space more narrowly V-shaped; upper incisors longer, but not as procumbent; molariform teeth smaller.

Topotypes of *moorei* can be distinguished from those of *Thomomys talpoides oquirrhensis* as follows: Size slightly larger; tail longer; ears larger, less pointed. Color: Lighter throughout; postauricular patches larger. Skull: More ridged and angular; nasals narrower posteriorly, but more dilated distally; posterior ends of nasals more deeply emarginate (while shallowly emarginate in *oquirrhensis*, they tend to be somewhat rounded); rostrum narrower; extension of premaxillae posterior to nasals greater; least interorbital breadth less; zygomatic arches more angular and widely spreading; zygomatic processes of maxillae heavier; interparietal smaller; tympanic bullae larger and more inflated ventrally; upper incisors generally longer.

The characters that distinguish *moorei* from *Thomomys talpoides parowanensis* are: Color: Lighter throughout. Skull: Broader, more angular and more nearly flat; zygomatic arches more widely spreading; zygomatic processes of maxillae heavier; posterior ends of nasals emarginate rather than rounded; upper incisors longer.

For comparisons of *moorei* with *Thomomys talpoides levis* and *wasatchensis* see accounts of these forms.

Remarks.—Specimens from Colton, show intergradation between *moorei*, *uinta* and *wasatchensis*, but are referable to *moorei* in the majority of characters. Specimens from Mount Nebo, and the mouth of Reddicks Canyon, in the Wasatch and San Pitch mountains, respectively, are intergrades between *moorei* and *wasatchensis*, but are referable to *moorei*.

That part of the Wasatch Mountains south of Spanish Fork Canyon is inhabited by pocket gophers that are intergrades between *moorei* and *wasatchensis*, but the cranial details show them to be referable to *moorei*. The range here ascribed to *moorei* consists of the Wasatch Plateau to the east of Sanpete Valley, the San Pitch Mountains and the southern part of the Wasatch Mountains. The type locality of *moorei* is situated in the southern end of a high valley that separates the Wasatch Plateau from the San Pitch and Wasatch mountains. Topotypical animals are larger and have more ridged, angular skulls than those from the mountains.

Specimens examined.—Total, 48, distributed as follows: *Utah County*: Near Payson Lake, 1 (R. H.); Mt. Nebo, 25 mi. SE Payson, 10,000 ft., 20; Colton, 8 (B. Y. U.). *Sanpete County*: 1 mi. S Fairview, 6,000 ft., 12 (U. S. N. M.). *Juab County*: Mouth of Reddicks Canyon, Wales Mountain (= San Pitch Mountains), 7,500 ft., 5. *Emery County*: Lake Creek, 11 mi. E Mt. Pleasant, 2 (C. M.).

Additional records.—*Sanpete County*: Ephraim, 5 (see Goldman, 1938:336).

***Thomomys talpoides fossor* Allen**

Thomomys fossor Allen, Bull. Amer. Mus. Nat. Hist., 5:51, April 28, 1893; Bailey, N. Amer. Fauna, 39:111, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):102, June, 1927; Hall, Univ. California Publ. Zool., 37:4, April 10, 1931.

Thomomys talpoides fossor Goldman, Journ. Mamm., 20:234, May 14, 1939.

Type.—Male, adult, skin and skull, No. 5240/4120, American Museum of Natural History; Florida, 7,200 ft., La Plata County, Colorado; June 25, 1892; collected by Charles P. Rowley (after Allen, type not seen).

Range.—In the mountains of San Juan and Grand counties, east of the Colorado and Green rivers.

Diagnosis.—Size medium (see measurements). Color: Upper parts Dresden Brown, grading over sides to Pale Buff on underparts; chin white; ears small, pointed, with deeply pigmented pinnae; postauricular patches grayish black; nose dusky. Skull: Long and narrow; nasals long, rounded proximally and usually simple distally; rostrum long; interparietal triangular; tympanic bullae large, and well inflated ventrally; basioccipital narrow; palate narrow; palatal pits shallow; dentition light.

Comparisons.—Near topotypes of *fossor* can be distinguished from topotypes of *Thomomys talpoides ocius* as follows: Size larger throughout. Color: Darker throughout, being dark brown as opposed to grayish. Skull: Longer and narrower; nasals and rostrum longer; extension of supraoccipital posterior to lambdoidal suture markedly less; tympanic bullae markedly smaller; upper incisors longer and not as strongly recurved.

Among the races of *Thomomys talpoides* occurring in Utah, *fossor* most closely resembles *Thomomys talpoides uinta* in color and size, but differs from it as follows: Ears smaller, more pointed and with more darkly pigmented pinnae. Skull: Longer, narrower and weaker; rostrum longer; nasals longer, and rounded proximally as opposed to markedly emarginate; interparietal triangular instead of roughly pentagonal; tympanic bullae larger and more inflated ventrally; basioccipital narrower; palate narrower, palatal pits shallower; dentition lighter.

Remarks.—Bailey (1915:111) remarked that *fossor* was one form that held its distinctive characters over a wide range. At that time, its range was understood to include practically all of the mountainous parts of Colorado, Utah as far west as the central part of the state, and parts of New Mexico, Arizona and Wyoming. Subsequently three new forms have been named from central Utah, (Goldman 1938:334-337) thereby showing variation to be much more prevalent than formerly supposed. The range of *fossor* in Utah, as now understood, is limited to the mountainous parts of the state south and east of the Colorado and Green rivers in Grand and San Juan counties.

The Utah specimens are not typical. At first glance some differences are noted in the premaxillae and nasals. Four specimens in the collections of the Museum of Natural History, University of Kansas, three from 3 miles east of Creede, Mineral County, and one from 10 miles east of Lake City, Hinsdale County, Colorado, both of which lie north and east of the type locality of *fossor* show the same characters as the Utah specimens.

Eight specimens from Oak Spring are intergrades between *fossor* and *ocius*. In size and color they are like *fossor*, but the skulls are intermediate. Because the animals are more like *fossor* in the majority of characters, they are here referred to that race.

As a result of these studies and due to the paucity of specimens from Utah, it is advisable, for the present, to refer all these Utah animals to *fossor*. Additional specimens may reveal characters that will merit the separation of the Utah animals from typical *fossor*; a desertlike area unfavorable to *Thomomys* exists between the type locality and eastern Utah.

Specimens examined.—Total, 21, distributed as follows: *Grand County*: Oak Spring, Middle Fork Willow Creek, 15 mi. N Thompson, 8 (C. M.); La Sal Mountains, 1 (U. S. N. M.); Warner Ranger Station, La Sal Mountains, 3 (B. Y. U.). *San Juan County*: Geyser Pass, 18 mi. SE Moab, La Sal Mountains, 3 (1, B. Y. U.; 2, C. M.); 5 mi. W Monticello, 1 (C. M.); Cooley Pass, 8 mi. W Monticello, 2 (C. M.); Joshua Flat, Elk Ridge, 8,300 ft., 3 (M. V. Z.).

***Thomomys talpoides parowanensis* Goldman**

Thomomys fossor parowanensis Goldman, Journ. Washington Acad. Sci., 28:334, July 15, 1938.

Thomomys talpoides parowanensis Goldman, Journ. Mamm., 20:234, May 14, 1939; Long, Journ. Mamm., 21:176, May 14, 1940.

Thomomys fossor Bailey, N. Amer. Fauna, 39:112, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):102, June, 1927; Hall, Univ. California Publ. Zoöl., 37:4, April 10, 1931; Presnall, Zion-Bryce Mus. Bull., 2:14, January, 1938; Tanner, Great Basin Nat., 1:111, 1940.

Type.—Male, adult, skin and skull, No. 158072, U. S. National Museum (Biological Surveys Collection); Brian Head, Parowan Mountains, 11,000 ft., Iron County, Utah; September 8, 1908; collected by W. H. Osgood; original number 3483 (after Goldman, type not seen).

Range.—High mountains of eastern Iron and Beaver counties, and western Kane and Garfield counties.

Diagnosis.—Size medium (see measurements). Color: Upper parts Sayal Brown moderately mixed with black, lightest on head; sides lightly washed with Buff; underparts Pinkish Buff, clearest on inguinal and pectoral regions; nose and cheeks dusky; postauricular patches large and black; front feet, hind feet and distal half of tail white. Skull: Long and fairly slender; zygomatic arches not widely spreading; nasals long; rostrum long and slender; posterior ends of nasals truncate or moderately emarginate; extension of premaxillae posterior to nasals usually short; tympanic bullae relatively small; upper incisors long and narrow; molariform teeth large.

Comparisons.—Compared with *Thomomys talpoides kaibabensis*, *parowanensis* differs as follows: Size smaller. Skull: Shorter; nasals shorter; zygomatic breadth less; nasals truncate or shallowly emarginate posteriorly as opposed to rounded; upper incisors narrower.

Topotypes of *parowanensis* differ from topotypes and near topotypes of *Thomomys talpoides uinta* as follows: Size larger. Color: Usually lighter; postauricular patches larger and darker; ears small with pinnae deeply pigmented as opposed to large and lightly pigmented. Skull: Larger; zygomatic arches more widely spreading; nasals longer; rostrum longer; posterior ends of nasals truncate or shallowly emarginate as opposed to deeply emarginate; sides of zygomatic arches nearly parallel and not so divergent posteriorly; interparietal larger and less quadrangular; extension of premaxillae posterior to nasals less; upper incisors less procumbent; molariform teeth larger.

Among named races of *Thomomys talpoides*, *parowanensis* most closely resembles *levis*, the race nearest geographically to the east, but differs from *levis* as follows: Size larger. Skull: Longer and wider; rostrum and nasals longer; interparietal quadrangular as opposed to roughly elliptical; upper incisors longer.

For comparisons with *Thomomys talpoides moorei* and *wasatchensis* see accounts of those forms.

Remarks.—The mountains of south central Utah are inhabited by pocket gophers that have been designated as *Thomomys talpoides parowanensis* and *T. t. levis* by Goldman (1938:334, 336). They are nearly indistinguishable in color and each is variable in cranial details. The diagnostic characters of each form occasionally appear, in varying degrees, throughout the range of the other. The Sevier River Valley separates the ranges ascribed to these two forms. This valley is inhabited by pocket gophers that belong to a different species, *Thomomys bottae*. The ranges of these two races of *talpoides* converge southward at the headwaters of the Sevier River. Specimens of *parowanensis* from the northern limits of its range from the Beaver Mountains in eastern Beaver County and those of *levis* from the northern limits of its range in the Fish Lake Mountains are readily distinguishable from each other. As the ranges converge to the southward, there is progressively more intergradation. The type locality of *parowanensis* is located in the southern part of its range, while that of *levis* is in the extreme northern part of its range. Therefore, due to the convergence of the two ranges at the south, the specimens from localities near the type locality of *parowanensis* show the greatest amount of intergradation, if we regard specimens of *parowanensis* from the type locality as typical of the race. Four specimens from Webster Flat, sixteen miles east of Cedar City, Iron County, and three from Duck Creek, Cedar Mountains, Kane County could equally well be assigned to either *levis* or *parowanensis*.

Specimens examined.—Total, 24, distributed as follows: *Beaver County*: Britts Meadows, Beaver Mountains, 8,500 ft., 7 (3, M. V. Z.; 2, U. S. N. M.; 2, C. M.); Puffer Lake, Beaver Mountains, 1 (U. S. N. M.); Kents Lake, Beaver Mountains, 1 (R. H.). *Iron County*: Lava Beds, 3-1/2 mi. SW Panquitch Lake, 1 (C. M.); Brian Head, Parowan Mountains, 2 (1, U. S. N. M.; 1, C. M.); Webster Flat, 16 mi. E Cedar City, 4; Bear Valley, 2 mi. E B. V. Ranger Station, 1 (R. H.). *Garfield County*: 1/4 mi. W Sunset Point, Bryce National Park, 8,000 ft., 1 (M. V. Z.). *Kane County*: Navajo Lake, 3 (R. H.); Duck Creek, Cedar Mountains, 9,000 ft., 3 (1, R. H.).

Additional records.—*Garfield County*: Panquitch Lake, 1 (see Goldman 1938:335). *Iron County*: Beaver Mountains, 9 (see Bailey, 1915:112); Buckskin Valley, 1 (see Goldman, 1938:335).

***Thomomys talpoides levis* Goldman**

Thomomys fossor levis Goldman, Journ. Washington Acad. Sci., 28:336, July 15, 1938.

Thomomys talpoides levis Goldman, Journ. Mamm., 20:234, May 14, 1939.

Thomomys fossor Bailey, N. Amer. Fauna, 39:112, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):102, June, 1927.

Type.—Female, adult, skin and skull, No. 158079, U. S. National Museum (Biological Surveys Collection); Seven Mile Flat, 5 mi. N Fish Lake, Fish Lake Plateau, 10,000 ft., Sevier County, Utah; October 1, 1908; collected by W. H. Osgood; original number 3616 (after Goldman, type not seen).

Range.—Fish Lake Mountains in Sevier County south into Garfield County, Utah.

Diagnosis.—Size medium (see measurements). Color: Upper parts near Sayal Brown, moderately mixed with black, darkest on head and middorsal region, grading to Cinnamon Buff on sides and flanks; underparts Pinkish Buff, clearest on inguinal and pectoral regions; chin, cheeks and nose dusky; postauricular patches large and black; front feet, hind feet and distal half of tail white; ears small and deeply pigmented. Skull: Slender and weak; zygomatic arches not widely spreading; posterior ends of nasals rounded; nasals moderately long and narrow; rostrum long and narrow; extension of premaxillae posterior to nasals short; interparietal usually much wider than long; pterygoid hamulae ridged; interpterygoid space usually narrowly V-shaped; upper incisors short.

Comparisons.—Compared with topotypes of *Thomomys talpoides moorei*, *levis* differs as follows: Size smaller; tail shorter. Color: Darker throughout, especially on dorsal surface due to more black of the underfur; underparts deeper buff. Skull: Narrower, less massive; zygomatic processes of maxillae weaker and not as widely spreading; interparietal generally wider; extension of premaxillae posterior to nasals less; posterior ends of nasals rounded rather than emarginate; upper incisors shorter, less procumbent.

Topotypes of *levis* differ from near topotypes of *Thomomys talpoides uinta* as follows: Size larger. Color: Upper parts slightly darker; postauricular patches much darker and larger; ears small and deeply pigmented as opposed to large and lightly pigmented; tail darker all around at base, with white part more extensive and with fewer buff-colored hairs. Skull: More convex dorsally; zygomatic arches more widely spreading and angular; nasals longer; rostrum longer; interparietal wider and more elliptical; posterior ends of nasals rounded as opposed to emarginate; extension of premaxillae posterior to nasals less; pterygoid hamulae more ridged; interpterygoid space more narrowly V-shaped; upper incisors shorter and less procumbent.

Topotypes of *levis* can be distinguished from those of *Thomomys talpoides kaibabensis* by markedly smaller measurements.

For comparisons with *Thomomys talpoides parowanensis* and *wasatchensis* see accounts of those forms.

Remarks.—Specimens from the Escalante Mountains and the Aquarius Plateau are not typical. They are of approximately the same color as *levis*, but are larger than *levis* and have cranial details that indicate intergradation with *kaibabensis* to the south. They resemble *kaibabensis* in large size, long nasals and widely spreading zygomatic arches, but are like *levis* in shape of the interparietal, extension of premaxillae posterior to the nasals, rounded posterior ends of nasals, ridged pterygoid hamulae and relatively short upper incisors. Additional material from these regions may prove these animals to merit separation and naming.

Specimens examined.—Total, 15, distributed as follows: *Sevier County*: Seven Mile Flat, 5 mi. N Fish Lake, Fish Lake Plateau, 10,000 ft., 2 (U. S. N. M.); Fish Lake Experiment Station, 2 (U. S. A. C). *Garfield County*: Posy Lake, Aquarius Plateau, 2 (B. Y. U.); 18 mi. N Escalante, 9,500 ft., 3; Steep Creek, Boulder-Teasdale Road, Boulder Mountain, 4 (B. Y. U.); Summit Birch Creek, Escalante Mountains, 2 (B. Y. U.).

MEASUREMENTS OF ADULT MALES OF THOMOMYS

(In millimeters)

Total length of tail	Length of hind foot	Basilar length of nasals	Length of zygomatic breadth	Mastoid breadth	Interorbital breadth	Alveolar length of upper	Extension of premax post. to nasals	Length of rostrum	Breadth of rostrum
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								molar series				
<i>T. t. gracilis</i> , 4; topotypes												
Av.	204	53	28	31.5	13.4	21.7	18.3	6.4	7.6	1.3	15.4	7.2
Min.	194	47	27	30.3	12.9	21.1	17.8	6.3	7.3	1.0	14.7	6.7
Max.	210	63	28	33.5	14.2	22.0	19.0	6.5	7.9	1.7	16.4	7.5
<i>T. t. oquirrhensis</i> , 4; topotypes												
Av.	209	58	28	32.2	13.9	21.9	19.0	6.9	7.6	0.9	15.8	7.7
Min.	197	55	28	31.9	13.7	21.4	18.5	6.7	7.2	0.6	15.5	7.5
Max.	216	60	29	32.8	14.3	22.8	19.5	7.1	7.9	1.0	16.2	7.9
<i>T. t. wasatchensis</i> , 10; topotypes												
Av.	221	67	28	31.3	13.4	21.5	18.9	6.5	7.4	1.1	15.1	7.4
Min.	204	60	26	27.4	11.6	19.1	17.2	6.0	6.6	0.9	14.0	6.7
Max.	237	75	31	34.5	15.2	23.7	20.4	7.3	8.0	2.0	16.5	8.2
<i>T. t. uinta</i> , 5; SW slope Bald Peak, Uinta Mts.												
Av.	199	51	27	31.5	13.1	21.7	19.4	6.3	7.6	1.1	15.2	7.4
Min.	185	47	26	29.6	12.1	20.3	19.0	5.7	7.3	0.7	13.5	7.2
Max.	208	54	28	32.8	13.8	22.2	20.0	6.5	7.8	1.4	15.6	7.6
<i>T. t. moorei</i> , 7; topotypes												
Av.	216	65	29	32.4	13.9	22.9	19.2	6.5	7.7	1.5	15.9	7.3
Min.	203	52	27	31.3	13.0	21.5	18.4	6.0	7.3	0.9	14.8	6.7
Max.	236	72	31	34.7	14.5	23.7	20.0	7.0	8.2	2.0	16.3	7.7
<i>T. t. fossor</i> , 8; Cascade Creek, La Plata Co., Colo.												
Av.	215	61	29	31.7	13.2	21.2	18.7	5.9	7.5	0.6	15.5	7.1
Min.	202	54	27	30.5	12.0	20.5	18.2	5.5	7.0	0.0	14.5	6.9
Max.	228	70	30	33.0	14.4	23.5	19.9	6.3	7.9	1.1	16.9	7.4
<i>T. t. ravus</i> , 3; topotypes												
Av.	248	73	30	35.2	14.6	24.8	21.4	6.3	8.3	2.4	17.1	8.2
Min.	244	70	29	34.5	14.3	23.6	20.5	6.0	8.2	2.2	16.7	8.1
Max.	253	74	30	35.9	15.1	25.7	22.5	6.7	8.4	2.7	17.5	8.5
No. 55270 (U. S. N. M.) <i>T. t. pygmaeus</i> , 1; topotype												
	165	40	20	24.6	10.2	16.3	15.1	5.4	5.9	0.7	12.0	5.7
No. 177506 (U. S. N. M.) <i>T. t. ocus</i> , 1; 12 mi. W Linwood, Henrys Fork, Wyo.												
	200	62	26	27.5	11.5	19.9	17.8	6.2	6.8	1.0	13.5	7.0
<i>T. t. parowanensis</i> , 2; Britts Meadow, Beaver Mountains												
Av.	215	59	28	34.3	14.5	22.4	18.6	6.0	8.1	1.4	17.3	7.9
Min.	202	48	27	34.1	14.1	22.0	18.4	5.8	8.0	1.0	17.2	7.6
Max.	228	69	29	34.6	14.8	22.7	18.9	6.2	8.2	1.7	17.3	8.2

MEASUREMENTS OF ADULT FEMALES OF THOMOMYS

(In millimeters)

Total Length	Length	Basilar	Length	Zygomatic	Mastoid	Interorbital	Alveolar	Extension	Length	Breadth	
length of tail	of hind foot	length	of nasals	breadth	breadth	breadth	length of upper molar series	of premax post. to nasals	of rostrum	of rostrum	
<i>T. t. gracilis</i> , 2; topotypes											
Av.	190	58	27	29.7	12.0	19.7	17.3	6.4	7.3	1.2	14.0 6.5

Min.	185	54	27	29.5	11.9	19.7	16.9	6.3	7.2	1.1	14.0	6.4
Max.	194	61	27	29.9	12.0	19.7	17.6	6.5	7.4	1.4	14.0	6.6
<i>T. t. oquirrhensis</i> , 7; topotypes												
Av.	203	56	27	30.2	12.9	20.4	18.2	6.8	7.5	0.8	14.8	7.2
Min.	193	52	25	28.5	12.2	19.5	17.5	6.6	6.7	0.5	14.2	6.9
Max.	215	59	28	31.5	13.3	21.0	19.1	7.2	8.0	1.0	15.5	7.5
<i>T. t. wasatchensis</i> , 19; topotypes												
Av.	205	62	27	31.5	12.7	20.5	18.0	6.5	7.4	0.9	14.6	7.2
Min.	180	52	23	28.1	11.2	19.3	17.2	6.2	6.0	0.6	13.0	6.8
Max.	222	70	30	32.5	14.5	22.0	19.9	6.7	8.1	1.2	16.2	7.5
<i>T. t. uinta</i> , 2; SW slope Bald Peak, Uinta Mts.												
Av.	181	49	25	28.4	11.6	19.8	17.3	6.6	7.2	1.3	13.5	6.8
Min.	177	47	25	28.3	11.6	19.8	17.2	6.4	7.0	1.1	13.3	6.8
Max.	185	50	25	28.4	11.6	19.8	17.4	6.7	7.3	1.5	13.6	6.8
<i>T. t. moorei</i> , 5; topotypes												
Av.	206	62	26	29.9	12.8	21.5	18.4	6.6	7.3	1.3	14.6	6.8
Min.	198	55	24	29.0	12.3	21.0	18.0	6.4	7.0	1.0	14.0	6.4
Max.	213	69	28	31.2	14.1	22.5	19.1	6.8	7.5	1.6	15.6	7.0
<i>T. t. fossor</i> , 4; Cascade Creek, La Plata Co., Colo.												
Av.	215	57	29	32.6	14.2	22.0	19.0	6.0	7.5	0.7	16.2	7.3
Min.	204	51	28	31.3	13.6	21.5	18.0	5.7	7.1	0.5	15.9	7.0
Max.	223	63	30	34.0	14.8	22.9	19.6	6.3	7.8	1.0	16.3	7.5
No. 13684 (C. M.) <i>T. t. ravus</i> , 1; topotype												
	241	71	28	35.7	14.5	24.4	21.5	6.2	7.8	2.7	17.1	8.1
No. 178868 (U. S. N. M.) <i>T. t. pygmaeus</i> , 1; Fossil, Wyo.												
	167	52	20	24.0	10.2	16.5	14.8	5.2	5.6	0.7	11.1	5.8
<i>T. t. ocious</i> , 3; topotypes												
Av.	201	60	25	30.0	13.5	20.5	17.9	6.2	7.2	0.8	15.0	7.4
Min.	196	57	25	29.9	13.0	19.9	17.5	6.1	7.1	0.5	14.7	7.3
Max.	205	64	25	30.1	14.0	21.5	18.6	6.3	7.3	1.0	15.3	7.5
<i>T. t. parowanensis</i> , 4; Britts Meadow, Beaver Mountains												
Av.	221	58	29	33.2	14.5	22.8	19.0	6.0	7.8	0.9	15.4	7.3
Min.	207	50	28	30.5	12.8	22.7	18.6	5.8	7.4	0.5	14.7	7.0
Max.	240	66	30	34.8	15.5	23.0	19.6	6.2	8.1	1.5	17.8	7.7
<i>T. t. levis</i> , 2; topotypes												
Av.	203	65	27	28.1	11.1	19.2	17.7	6.1	6.9	0.8	13.0	6.8
Min.	199	61	26	28.0	10.6	18.9	17.5	5.8	6.6	0.6	12.8	6.6
Max.	206	70	27	28.2	11.6	19.5	17.9	6.4	7.2	1.0	13.2	7.0

Thomomys bottae (Eydoux and Gervais)

Thomomys bottae is a southern species that, within the Great Basin, reaches the most northern limits of its distribution in Utah. The animals of this species inhabit the lower valleys, and with the exception of the Oquirrh Mountains, inhabit also the mountains in that part of the state west of the central mountain ranges. The specific characters are: Sphenorbital fissure present; incisive foramina posterior to infraorbital canal; anterior prism of P4 rounded; interparietal relatively small; lambdoidal suture straight in region of interparietal, in Utah specimens.

Thomomys bottae aureiventris Hall

Thomomys perpallidus aureiventris Hall, Univ. California Publ. Zoöl., 32:444, July 8, 1930; Univ. California Publ. Zoöl., 37:3, April 10, 1931.

Thomomys bottae aureiventris Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935.

Type.—Male, adult, skin and skull, No. 43980, Museum of Vertebrate Zoölogy, University of California; Fehlman Ranch, 3 mi. N Kelton, 4,225 ft., Box Elder County, Utah; September 27, 1929; collected by Louise Kellogg; original number 451.

Range.—Northwestern Utah, and extreme western Utah as far south as the southern end of the Deep Creek Mountains.

Diagnosis.—Size medium (see measurements); claws on front feet small. Color: Near Cinnamon on dorsal and ventral surfaces; inguinal region, front and hind feet and distal third to half of tail white; nose, cheeks and postauricular patches grayish black. Skull: Moderately angular and ridged; zygomatic arches nearly parallel with sides of skull; jugals vertical; marked thickening at union of jugal and zygomatic process of maxilla; greatest zygomatic breadth at anterior part of arches; interpterygoid space lyre-shaped; ventral margin of jugal concave dorsally; nasals long and denticulate distally; parietal ridges bowed in at two places, at coronal suture and at middle of interparietal; paroccipital processes extremely well developed; dorsal frontomaxillary suture usually straight.

Comparisons.—From near topotypes of *Thomomys bottae centralis*, *aureiventris* differs as follows: Size larger; tail shorter; hind foot longer; claws on front feet shorter. Color: Slightly darker on upper parts, but with greater extension of white on ventral surface. Skull: Zygomatic breadth greater; greatest width across zygomatic arches at anterior rather than posterior region; zygomatic arches thicker at union of jugals and zygomatic processes of maxillae; dorsal frontomaxillary suture less convex medially; mastoid breadth greater; extension of premaxillae posterior to nasals less; interpterygoid space lyre-shaped rather than V-shaped.

From topotypes of *Thomomys bottae albicaudatus*, *aureiventris* can be distinguished by: Size larger; hind foot longer. Color: Markedly lighter throughout, Cinnamon as opposed to near (13"n) Black. Skull: Larger in all but three measurements taken; extension of premaxillae posterior to nasals less; alveolar length of upper molar series shorter; zygomatic arches widest anteriorly rather than posteriorly; thickening at union of jugal and zygomatic process of maxilla markedly greater; interpterygoid space lyre-shaped as opposed to V-shaped; lacrimal processes more globose at tips.

Thomomys bottae aureiventris can be readily distinguished from *T. b. bonnevilliei*, *sevieri*, *wahwahensis*, and *convexus* by larger size in all measurements taken and darker coloration. The same differences obtain in comparison with *T. b. tivius* and *stansburyi* except that *aureiventris* is much lighter colored. See comparisons under those forms.

Remarks.—*T. b. aureiventris* has one of the most extensive ranges of any race of *T. bottae* occurring in Utah. The range extends from the valleys of the northwest corner of the state south along the extreme western margin of the state approximately to the southern end of the Deep Creek Mountains. This ascribed range practically bounds the northwest and western margins of the great salt desert in Box Elder and Tooele counties. As far as known, this great waste area harbors no members of the Geomyidae. Pocket gophers were available from four localities in addition to the type locality. In these four localities all of the animals were intergrades. The three specimens from Queen of Sheba Canyon, Deep Creek Mountains, although smaller than *aureiventris* in every measurement taken, resemble it in color and general configuration of the skull. The animals from Trout Creek and Ibapah at the southern end of the range, although referred to *aureiventris*, are intermediate between it and *centralis*. In color and measurements they more closely resemble *centralis*, but the skulls closely resemble those of *aureiventris*. The skulls show some slight characteristics of *bonnevilliei*, the form to the east, which indicate an early relationship between the two. Specimens from the east side of Tecoma Range, adjacent to Pilot Peak, although referred to *aureiventris* are intergrades between it and *centralis*. Although this locality is nearer the type locality of *aureiventris* than any of the other record stations, the animals show the maximum departure from topotypes in morphological features. In color they approach *centralis*, and agree with it in one-half of the measured characters. The general configuration of the skull and a majority of the critical diagnostic characters, for example, jugal thickening, are more nearly as in *aureiventris*. From the above remarks it is readily understood that this subspecies is extremely variable.

Specimens examined.—Total, 55, distributed as follows: *Box Elder County*: Fehlman Ranch, 3 mi. N Kelton, 4,255 ft., 8 (7, M. V. Z.); Utah-Nevada Boundary, E Side Tecoma Range, 4,300 ft., 12. *Tooele County*: Ibapah, 5,000 ft., 21. *Juab County*: Queen of Sheba Canyon, W side Deep Creek Mountains, 5,600 ft., 11.

Thomomys bottae robustus new subspecies

Type.—Male, adult, skin and skull, No. 2726, Museum of Zoölogy, University of Utah; Orr's Ranch, Skull Valley, 4,300 ft., Tooele County, Utah; June 19, 1938; collected by S. D. Durrant; original number 1583.

Range.—Skull Valley, Tooele County, Utah.

Diagnosis.—Size medium (see measurements); tail short; hind foot short. Color: In a series of 24 animals, upper parts vary from Pale Smoke Gray (4 specimens) through Cinnamon Buff (19 specimens) to Dark Mouse Gray (1 specimen). The Cinnamon Buff color is considered to be typical. Color grading to lighter on underparts; postauricular patches small and grayish black; front and hind feet and distal part of tail white. Skull: Small, flat and heavily ridged; nasals short; zygomatic arches heavy and widely spreading, widest posteriorly at union of jugal and squamosal; union of jugal and zygomatic process of maxilla thickened, with a ventrally directed spinous process in sixty percent of the specimens; occasionally there is a second process, also directed ventrally at union of jugal and zygomatic process of squamosal; zygomatic arches convex dorsally; deep dorsal depression present in frontal bones in mature specimens; lacrimal processes prominent, projecting well above the arch at the anteromedial angle of the orbit; interpterygoid spaces V-shaped; tympanic bullae well inflated ventrally; upper incisors short, and pale; when placed on a flat plane the dorsal surface of the skull is nearly parallel to the substratum; space enclosed within the zygomatic arches nearly quadrangular.

Comparisons.—From topotypes of *Thomomys bottae aureiventris*, *robustus* can be distinguished as follows: Size smaller; tail and hind foot shorter. Color: Lighter throughout. Skull: Smaller, more heavily ridged and more nearly flat; nasals shorter; rostrum relatively wider and shorter; zygomatic arches shorter and relatively more widely spreading with greatest width posteriorly as opposed to anteriorly; junction of jugal and zygomatic process of maxilla not as prominent; *aureiventris* shows no spinous process at this junction; lacrimal processes larger and projecting farther dorsally; enclosed space within zygomatic arches roughly quadrangular as opposed to triangular; mastoid part of tympanic bullae less exposed; sphenorbital fissure smaller; interpterygoid space V-shaped rather than lyre-shaped; palatal pits smaller and shallower; tympanic bullae smaller, but more inflated ventrally; basioccipital averaging relatively wider; molars smaller; upper incisors shorter, smaller and cadmium yellow as opposed to orange yellow.

Comparisons of *robustus* with topotypes of *Thomomys bottae albicaudatus* show the following: Size smaller. Color: Lighter throughout; postauricular patches smaller and lighter. Skull: Smaller, more compact and more nearly flat; rostrum shorter and more nearly straight; lacrimal processes larger, projecting higher above the anteromedial angle of the orbit; parietal ridges uniformly heavier; mastoid width actually as well as relatively wider; zygomatic arches heavier and relatively much wider (males 76.2 percent of basilar length, females 73.8 percent as opposed to males 73.8 percent and females 73.5 percent); union of jugal and zygomatic process of maxilla uniformly more thickened; spinous process at jugal-maxillary suture present; zygomatic arches much more concave on ventral surface; uniform deep depression present in mature adults, between frontal processes of premaxillae, and anterior interorbital region of frontals; extension of premaxillae posterior to nasals less; sphenorbital fissure more constricted; tympanic bullae more inflated ventrally, extending well ventrad of basioccipital; palatal pits shallower and smaller; molars smaller; upper incisors shorter, narrower and paler (see comparison of *aureiventris*).

From near topotypes of *Thomomys bottae centralis* from 1 mile east of Garrison, Millard County, Utah, *robustus* differs in: Size smaller; tail and hind foot shorter. Color: Lighter, terminal bands of hair cinnamon, but because more black in underfur the animals appear darker; postauricular patches smaller and lighter. Skull: Shorter, more nearly flat and much more heavily ridged; nasals shorter; rostrum shorter and wider; lacrimal processes larger and projecting higher above anteromedial angle of orbit; zygomatic arches heavier, shorter, more angular and actually as well as relatively wider; jugals thicker; angle between maxillary plate and rostrum less obtuse; spinous process at jugal-maxillary suture present; extension of premaxillae posterior to nasals less; parietal ridges much more pronounced; looked at from above, space enclosed within zygomatic arches more quadrangular in shape as opposed to roughly triangular; tympanic bullae more inflated ventrally; molars smaller; upper incisors shorter, narrower and paler.

The characters that distinguish *robustus* from topotypes of *Thomomys bottae wahwahensis* are: Size slightly smaller. Color: Darker throughout. Skull: Rostrum longer and narrower; nasals longer; zygomatic arches wider and longer; lacrimal processes larger and projecting higher above anteromedial angle of the orbit; parietal ridges more

roughened; tympanic bullae much larger and more inflated ventrally; supraoccipital higher; middorsal depression in frontals present. For comparisons with *Thomomys bottae bonnevillae* see account of that form.

The remaining forms from the Bonneville Basin, namely, *Thomomys bottae sevieri*, *convexus*, *tivius* and *stansburyi* are all easily distinguished from *robustus*. Specimens of *sevieri* are paler, smaller in every measurement taken, and the skulls are weaker and less angular. All specimens of *convexus* are paler, the skulls are more convex dorsally and narrower, with less ridging and angularity. Both *tivius* and *stansburyi* are small dark forms, with weak, smooth, small skulls as compared with *robustus* which is light colored and has compact, ridged and angular skulls.

Remarks.—Twenty-three specimens were obtained at a small isolated spring. Critical study of animals taken only a few miles to the east prove them to be so different as to be referable to another subspecies, *albicaudatus*. *T. b. robustus* is an endemic form in this desert valley. The variable color is noteworthy but difficult to explain in an isolated population as small as this one. All five of the gray animals are females of which four are lactating adults. The affinities of this subspecies are with *albicaudatus* to the east, but enough time has elapsed since isolation to enable them to differentiate.

Specimens examined.—Total, 23, from the type locality.

***Thomomys bottae minimus* Durrant**

Thomomys bottae minimus Durrant, Proc. Biol. Soc. Washington, 52:161, October 11, 1939; Marshall, Journ. Mamm., 21:154, May 14, 1940.

Type.—Male, adult, skin and skull, No. 263942, U. S. National Museum (Biological Surveys Collection); Stansbury Island, Great Salt Lake, Tooele County, Utah; June 25, 1938; collected by William H. Marshall; original number 141.

Range.—Known only from the type locality.

Diagnosis.—Size small (see measurements); tail relatively long. Color: Upper parts Pinkish Buff, darker on head; underparts Pale Pinkish Buff; front and hind feet white; nose, chin and postauricular patches black. Skull: Long, slender and nearly devoid of ridges; braincase moderately inflated; interparietal quadrangular; zygomatic arches weak, widest in temporal region, but neither widely spreading nor angular; nasals straight and truncate posteriorly; extension of premaxillae posterior to nasals relatively great; tympanic bullae moderately inflated; palatal pits deep; rostrum short but narrow; interpterygoid space moderately lyre-shaped; upper incisors narrow; molars light.

Comparisons.—Compared with topotypes of *Thomomys bottae albicaudatus*, *minimus* differs as follows: Size markedly smaller; claws on front feet shorter and weaker. Color: Markedly lighter throughout, being Pinkish Buff as contrasted with near (13"n) Black. Skull: Smaller in every measurement taken; slender, smooth, weak and nonangular as opposed to ridged, robust, wide and angular; zygomatic arches much weaker and not so widely spreading posteriorly; ascending processes of premaxillae much narrower; extension of premaxillae posterior to nasals less; interpterygoid space moderately lyre-shaped as opposed to V-shaped; dentition lighter.

Topotypes of *minimus* differ from those of *Thomomys bottae aureiventris* as follows: Size markedly smaller. Color: Lighter dorsally and no "gold color" on underparts. Skull: Markedly smaller in every measurement taken; weak, smooth and slender as opposed to ridged, angular and robust; zygomatic arches weak and widest posteriorly rather than heavy and widest anteriorly; no great thickening at region of union of jugal and zygomatic process of the maxilla; jugals more nearly straight rather than concave laterally; interpterygoid space not so markedly lyre-shaped; dentition lighter.

The races nearest geographically to *minimus* are *Thomomys bottae nesophilus* and *T. b. stansburyi*. For comparisons see accounts of those forms.

Remarks.—This subspecies is the smallest of all the races of *Thomomys bottae* occurring in Utah. As far as known it is endemic to Stansbury Island, and since the Pleistocene Lake Bonneville attained its highest level has remained on that part of Stansbury Island that was above this high level. (See comments under *nesophilus*.) The sandy nature of the soil and the desert conditions of the area that has since been exposed at lower levels apparently do not constitute a favorable environment. Unlike *nesophilus* from Antelope Island, this form does not have its affinities with *albicaudatus*, the valley form of the adjacent mainland, but does show affinities with *stansburyi*, the nearest

mountain form on the mainland. This is easily understood when one realizes that Stansbury Island is only an isolated part of Stansbury Mountain that projects northward as a peninsula into Great Salt Lake. The history of Stansbury Island with reference to isolation of *minimus* parallels that of *nesophilus* on Antelope Island. See discussion under *nesophilus*.

Specimens examined.—Total, 5, as follows: *Tooele County*: Stansbury Island, Great Salt Lake, 5 (U. S. N. M.).

***Thomomys bottae nesophilus* Durrant**

Thomomys bottae nesophilus Durrant, Bull. Univ. Utah, 27 (No. 2):2, October, 1936; Marshall, Journ. Mamm., 21:156, May 14, 1940.

Type.—Male, adult, skin and skull, No. 1136, Museum of Zoölogy, University of Utah; Antelope Island, Great Salt Lake, Davis County, Utah; April 20, 1935; collected by S. D. Durrant; original number 761.

Range.—Known only from the type locality.

Diagnosis.—Size medium (see measurements); claws on front feet long. Color: Upper parts Cinnamon Buff; lighter below; sides Pinkish Buff interspersed with gray; pectoral and inguinal regions Cinnamon; nose grayish black; postauricular patches black. Skull: Interparietal wedge-shaped; tympanic bullae small; dorsal surface of lambdoidal prominence 3 mm. wide rather than developed as a crest; jugals nearly straight; zygomatic arches strongly rectangular.

Comparisons.—Compared with topotypes of *Thomomys bottae albicaudatus*, *nesophilus* is of approximately the same size, but differs as follows: Claws on front feet longer. Color: Lighter throughout; tail white terminally, but much darker at base; postauricular patches smaller. Skull: Interparietal wedge-shaped as opposed to roughly quadrangular; lambdoidal eminence more of a crest than a ridge; tympanic bullae smaller; jugals more nearly straight; zygomatic arches more nearly rectangular.

From topotypes of *Thomomys bottae aureiventris*, *nesophilus* differs in: Size smaller; claws on front feet longer. Color: Darker throughout; postauricular patches larger. Skull: Heavier, more massive; zygomatic arches more robust and convex laterally rather than concave; interparietal wedge-shaped rather than roughly quadrangular; braincase more nearly flat; tympanic bullae markedly smaller; upper molariform series longer; molariform teeth wider and heavier; interpterygoid space V-shaped rather than lyre-shaped.

The race nearest geographically to *nesophilus* is *T. b. minimus* from Stansbury Island, Great Salt Lake. It can easily be distinguished from *minimus* by the following features: Size much larger; claws on front feet longer and thicker. Color: Darker throughout; postauricular patches larger and with more admixture of buff colored hairs. Skull: Larger in every measurement taken; wide and robust as opposed to narrow and slender; zygomatic arches more widely spreading and angular; braincase more nearly flat; tympanic bullae actually larger, but relatively smaller; lambdoidal eminence flat-topped rather than a crest; interparietal wedge-shaped as opposed to quadrangular; teeth larger.

Remarks.—The affinities of *nesophilus* of Antelope Island are unquestionably with *albicaudatus* of the eastern and southern mainland. At the time of this writing (1945), Antelope Island is not truly an island, but only the tip of a broad peninsula projecting westward into Great Salt Lake. Nevertheless, the area of occurrence of *nesophilus* is effectively isolated by the exposed, sandy lake bottom that is unsuited to occupancy by pocket gophers. Fluctuations in the level of the Great Salt Lake have broken and reestablished this connection with the mainland many times. Each of the several other kinds of mammals which are known from both the island and the mainland show no differentiation on the island. These are kinds (see Marshall, 1940:156), which more freely cross the exposed, sandy lake bottom. I, myself, have noted tracks of coyotes going to and from the island. The pocket gopher, *nesophilus*, so far as known is the only mammal which has developed a subspecies endemic to the island. The beach levels of Pleistocene Lake Bonneville are well marked on both Antelope Island and Stansbury Island, which is fifteen miles west of Antelope Island. On the eastern side of Antelope Island the lower beach levels of this prehistoric lake are farmed. Although sought for elsewhere on this island, pocket gophers were found only in the farmed land. On Stansbury Island there has been no farming, and the endemic pocket gophers, *minimus*, although sought for elsewhere on that island were found only above the highest beach levels of the ancient lake. Evidently these pocket gophers still occupy only that part of Stansbury Island that projected above water during the greatest height of Lake Bonneville. Farming on Antelope Island may have developed a more favorable environment for

pocket gophers, thus causing them to move down to the lower levels from that part of the island that was above water during Pleistocene times.

Specimens examined.—Total, 5, from the type locality.

***Thomomys bottae stansburyi* new subspecies**

Type.—Female, adult, skin and skull, No. 2045, Museum of Zoölogy, University of Utah; South Willow Creek, Stansbury Mountains, 7,500 ft., Tooele County, Utah; July 2, 1937; collected by O. S. Walsh and S. D. Durrant; original number 1257 of Durrant.

Range.—Stansbury Mountains, Tooele County, Utah.

Diagnosis.—Size small (see measurements). Color: Upper parts Saccardo's Umber, darker on head; sides and underparts Pinkish Buff; nose, chin and postauricular patches black; front and hind feet and distal third to half of tail white. Skull: Small, slender, weak and smooth; zygomatic arches light and not widely spreading; zygomatic arches actually as well as relatively short; interparietal generally quadrangular; nasals relatively long and slender; interpterygoid space narrowly V-shaped; basioccipital fairly wide; tympanic bullae moderately inflated ventrally; dentition light.

Comparisons.—Topotypical specimens of *stansburyi* can be readily distinguished from those of *Thomomys bottae centralis*, *aureiventris* and *albicaudatus* by being smaller in every measurement taken, particularly those of the skull; the skull is weaker and smoother. In color *stansburyi* is like *albicaudatus* but is much darker throughout than *aureiventris* and *centralis*.

Comparisons of topotypes of *stansburyi* with those of *Thomomys bottae sevieri* show them to be of approximately the same size, but to differ as follows: Color: Darker throughout. Skull: Zygomatic arches shorter; tympanic bullae less inflated ventrally; zygomatic breadth less; mastoid breadth greater; width across alveolar processes of maxillae greater; alveolar length of upper molar series greater; molariform teeth larger.

Compared with topotypes of *Thomomys bottae minimus*, *stansburyi* is seen to be of larger size and darker color throughout, with a skull that is larger in most every measurement taken, although of the same slender, smooth, nonangular type.

Among named races of *Thomomys bottae*, *stansburyi* most closely resembles *tivius*, a small, dark, mountain form from central Utah. Size and color are almost the same but *stansburyi* differs in: Tail shorter; hind foot averaging slightly longer. Skull: Generally larger in every measurement taken; zygomatic arches shorter; width across alveolar processes of maxillae greater; zygomatic arches more widely spreading, and widest in extreme posterior region rather than in region of jugal-squamosal suture.

Remarks.—The Stansbury Mountains are separated from the Oquirrh Mountains by the Stockton Bar, and from the Onaqui Mountains, which are in reality a continuation of the Stansbury Mountains, by only a low pass. Pocket gophers from Clover Creek, Onaqui Mountains and Little Valley, Sheeprock Mountains, although intergrades between *robustus* and *albicaudatus* are dark in color like *stansburyi*. These intergrades are large, dark colored, and have heavy, ridged, angular skulls. It appears that *stansburyi* is a mountain subspecies derived from *albicaudatus* of the valley. It would be instructive to artificially transplant gophers from mountains to valleys, and *vice versa*, so as to reveal what effects if any on the animals' morphology the environment might have in one or a few generations. Gophers are well known to be very plastic, and such an experiment as suggested might call for modification of the view, held here, that the differential features of gophers from South Willow Creek and, say, Bauer, are hereditary.

Specimens examined.—Total, 11, from the type locality.

***Thomomys bottae albicaudatus* Hall**

Thomomys perpallidus albicaudatus Hall, Univ. California Publ. Zoöl., 32:444, July 8, 1930; Univ. California Publ. Zoöl., 37:3, April 10, 1931.

Thomomys bottae albicaudatus Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935; Durrant, Bull. Univ. Utah, 28 (No. 4):5, August 18, 1937.

Thomomys perpallidus aureiventris Hall, Univ. California Publ. Zoöl., 37:3, April 10, 1931.

Type.—Male, adult, skin and skull, No. 43971, Museum of Vertebrate Zoölogy, University of California; Provo, 4,510 ft., Utah County, Utah; October 17, 1929; collected by Annie M. Alexander; original number 506.

Range.—From the area between the Great Salt Lake and the Wasatch Mountains south along the western margin of the central mountains of the state to the Sevier River, in Juab County, west into Tooele County to the Onaqui and Sheeprock mountains.

Diagnosis.—Size medium (see measurements); claws on front feet medium. Color: Upper parts near (13""n) Black, grading over sides and flanks to Pinkish Cinnamon on underparts; chin, nose, top of head and postauricular patches black; front feet, hind feet and distal third to half of tail white. Skull: Angular and ridged; zygomatic arches moderately wide spreading, widest posteriorly; paroccipital processes weak; zygomatic processes of maxillae convex anteriorly; lacrimal processes small and peglike; jugals convex dorsally on ventral surface; nasals short, rounded distally and truncate proximally; parietal crests bowed in, in two places; interpterygoid space broadly V-shaped.

Comparisons.—For comparisons of *albicaudatus* with *Thomomys bottae aureiventris* and *centralis* see accounts of those forms.

Topotypes of *albicaudatus* are dark colored and can be distinguished from those of *Thomomys bottae birdseyei*, *tivius*, *stansburyi* and *contractus* which are also dark forms, by larger size and larger, more robust skulls (see accounts of those forms). It can be distinguished from the remainder of the known subspecies of *Thomomys bottae* in Utah by darker color and by cranial details (see accounts of those forms).

Remarks.—The range of *albicaudatus* is larger than that of any other race of *Thomomys bottae* limited to Utah. Specimens are available from thirty localities which represent widely varied habitats and environments. This subspecies consists of many highly variable local populations, and the marginal populations intergrade freely with adjacent races. In many populations, it is really difficult to recognize the relationships on account of the great variation, and one is frequently tempted to name some of them as distinct. Careful study of the large number of specimens has enabled me to recognize diagnostic characters common to all of these variable populations. The animals range from large and dark at the north to small and light at the south.

The Jordan River bisects Salt Lake County from north to south. Pocket gophers were taken at nine places east of the river, and at three places west of it.

Gophers from Salt Lake City and environs (east of the river) vary in color from almost black to dark cinnamon. Specimens from Draper, which locality is likewise east of the river, are uniformly lighter, but also vary in color. The skulls of animals from both localities are indistinguishable from each other and closely resemble those of topotypes. Specimens from the west side of the river, from Riverton, two miles west of Murray and Rose Canyon, Oquirrh Mountains, all are lighter in color than topotypes. The color varies from darkest at the north at Murray to lightest at the south at Riverton. This is exactly the reverse of what would be expected since Riverton is the locality geographically nearest to the type locality, Provo. The skulls are quite uniform and are all referable to *albicaudatus*. The Jordan River may be one factor which causes this lack of uniformity between the animals from the two sides of the river. Davis (1939:56-57) states that rivers are not barriers to movement of pocket gophers where the river completely freezes over and has the ice covered with thick snow. Although the Jordan River does occasionally freeze over, it is never frozen for more than a few days at a time, and snow in this area does not last for long periods. The material at hand indicates that the gophers from both sides of the river are referable to the same subspecies *albicaudatus*. The animals from the east side of the river are in the aggregate of characters the most typical of *albicaudatus* of any in the entire range. Those from the west side of the river, although definitely referable to *albicaudatus* do show some intergradation with *Thomomys bottae robustus*, the subspecies to the west.

The specimens from Bauer, Tooele County, are relatively uniform in color, and are considerably lighter than topotypes of *albicaudatus*. Their upper parts vary from Sepia to Saccardo's Umber as compared with near (13""n) Black of the topotypes. The sides and underparts are lighter, due primarily to much less black in the underfur. They average slightly longer in total length, but shorter in hind foot. All cranial measurements are slightly smaller than

in topotypes of *albicaudatus*. The shape of the skull closely resembles that of *albicaudatus*, although the rostrum, nasals, upper incisors and posterior tongues of the premaxillae tend to be narrower. This narrowness indicates intergradation with *Thomomys bottae stansburyi*, the race nearest to the west. These animals are in the majority of characters referable to *albicaudatus*.

Bauer is situated in extreme western Tooele Valley at the foot of Stockton Bar, a low pass between the Stansbury and the Oquirrh mountains. This valley lies to the west of the aforementioned Jordan River. Although these gophers are definitely referable to *albicaudatus* they are more unlike topotypes than are the animals from Riverton.

The specimens from Settlement Canyon, Oquirrh Mountains, Tooele County, show the same characteristics as those from Bauer.

In a large series of animals from St. John, in Rush Valley, Tooele County, the upper parts vary from black, even darker than topotypes of *albicaudatus*, to Tawny Olive, and the underparts vary from black through Cinnamon Buff to Pinkish Buff. Most of the animals are Cinnamon Buff. Although variable they approach *albicaudatus* in color. The total length, tail and hind foot of males are longer than in topotypes of *albicaudatus*; females differ in the same direction but only slightly. In both sexes the zygomatic breadth is less, but the mastoid breadth is greater than in *albicaudatus*. In size and shape of the lacrimal processes, and the great thickening of the jugal at the maxillo-jugal suture they approach *robustus*. They are much larger, however, and in the majority of characters are referable to *albicaudatus*.

What has just been said relative to the animals from St. John applies also to those from Clover Creek in the Onaqui Mountains of Tooele County. At the latter locality the tendencies towards *robustus* are accentuated. This is to be expected, since this locality is midway between St. John and the type locality of *robustus*. All characters considered, these animals are all referable to *albicaudatus*.

The animals from Little Valley, Sheeprock Mountains, Tooele County, resemble *albicaudatus* in color. They vary on the upper parts from near (1) Sepia to Clay Color, and ventrally from nearly black to Pinkish Buff. They are markedly smaller in every measurement taken, except zygomatic and mastoidal breadths, and extension of premaxillae posterior to nasals. This relatively greater breadth indicates intergradation with *robustus* to the west. These gophers are smaller in most measurements than any other population referred to *albicaudatus*. This is understandable because gophers from mountains usually are smaller and have weaker, smoother skulls than animals from low lands. Although approaching *robustus* in size and in some aforementioned cranial details, the aggregate of characters including color, make these animals referable to *albicaudatus*.

The animals from Fairfield, Utah County, are closer geographically to the type locality of *albicaudatus* than any other series, but morphologically are the least like topotypes. At first glance one is struck with the differences. They are uniformly Clay Color above, with Cinnamon Buff sides and flanks and Pinkish Buff underparts. Their color closely approaches that of *robustus* to the west which has Cinnamon Buff on the upper parts. Examination of eleven measurements of males and the same number for females, shows that the animals are nearest to *robustus* in two measurements, to *albicaudatus* in 12, distinct in 7 and intermediate in one. The general appearance of the skull is intermediate between that of the two above mentioned forms. The differences from *albicaudatus* in size and color may be correlated with the differences in soil at Fairfield and Provo. At Fairfield the soil is light-colored clay, but at Provo it is sandy and darker. Although they are intergrades between *robustus* and *albicaudatus*, the animals are referred to the latter race. Utah Lake and its outlet, the Jordan River, make a partial barrier between populations at Fairfield and at the type locality at Provo. During Pleistocene times, when Lake Bonneville was present it formed a complete barrier. Enough time has evidently elapsed since the disappearance of this lake to allow *albicaudatus*, the mainland form, to expand its range to the west. Intergradation has taken place, with the result that the animals from Fairfield, although unstable, agree with the mainland form, *albicaudatus*, in a majority of their characters.

Pocket gophers were taken at four localities from north to south in eastern Juab County. They range in color from Ochraceous Tawny on the upper parts and Cinnamon Buff on the underparts to shades that are slightly lighter. All are much lighter than topotypes of *albicaudatus*. The general configuration of the skull is the same as that of *albicaudatus*, and this is especially true in the females. In the narrower rostrum and weaker dentition they approach *contractus*, but are distinctly lighter colored. Hall (1931:3) referred one specimen from Nephi, Juab County, to *Thomomys bottae aureiventris*. Since that time *Thomomys bottae lenis* which has some affinities with *aureiventris* has been described (see account of *contractus*). The large series now available from Nephi and nearby

localities do show some intergradation with *lenis*, in that four characters are more as in *lenis* and *contractus* and seven characters are more as in *albicaudatus*. Although differing markedly in many respects from topotypes of *albicaudatus* they fit the aforementioned concept of this subspecies, and are being treated as a variable local population of it.

Provo is the locality listed for specimens which were available to naturalists from 1875-1877. To these specimens the following names were applied: *Thomomys talpoides bulbivorus* Coues (1875:256; 1877:627) and *Thomomys talpoides umbrinus* Coues and Yarrow (1875:112). Possibly these names were applied to the animals currently known as *Thomomys bottae albicaudatus* which does occur at Provo. Without the opportunity to examine the actual specimens, which so far as I know are no longer in existence, I cannot exclude the possibility that the locality designation "Provo" was used in a general sense to include pocket gophers taken a few miles to the eastward of Provo, where it is known that pocket gophers of only the species *Thomomys talpoides* (current terminology) occur.

Specimens examined.—Total, 239, distributed as follows: *Davis County*: Bountiful, 4,500 ft., 1. *Salt Lake County*: Salt Lake City and environs, 4,300 ft., 51; 2 mi. W Murray, 4,300 ft., 6; Riverton, 4,300 ft., 11; Draper, 4,500 ft., 7; Rose Canyon, Oquirrh Mountains, 5,650 ft., 4. *Tooele County*: Bauer, 4,500 ft., 30; Settlement Creek, Oquirrh Mountains, 6,500 ft., 1; St. John, 4,300 ft., 28; Clover Creek, Onaqui Mountains, 5,500 ft., 15; Vernon, 4,300 ft., 2 (U. S. A. C.); Little Valley, Sheeprock Mountains, 5,500 ft., 20. *Utah County*: Fairfield, 4,800 ft., 24; Provo, 4,400 ft., 20 (8, B. Y. U.; 12, M. V. Z.). *Juab County*: Neff Farm, 4 mi. N Nephi, 5,000 ft., 2 (1, R. H.); Nephi, 5,000 ft., 1 (M. V. Z.); 2 mi. S Nephi, 4,700 ft., 14; 7 mi. SW Nephi, 6,000 ft., 2.

***Thomomys bottae bonnevilliei* new subspecies**

Type.—Male, adult, skin and skull, No. 3576, Museum of Zoölogy, University of Utah; Fish Springs, 4,400 ft., Juab County, Utah; June 8, 1940; collected by S. D. Durrant; original number 1955.

Range.—Known only from the type locality.

Diagnosis.—Size medium (see measurements); claws on front feet small. Color: Entire dorsal surface Warm Buff; sides near (e) Cinnamon Buff, underparts near (16") Pale Pinkish Buff; inguinal region, front and hind feet and distal part of tail white; top of head, nose and cheeks grayish black; postauricular patches small and grayish black; ears small, pointed and with heavily pigmented pinnae. Skull: Angular, short and wide; nasals of medium length, narrow proximally but widely flared distally; interparietal small; lambdoidal suture concave towards the interparietal; zygomatic arches uniformly widely spreading; interpterygoid space widely V-shaped; extension of premaxillae posterior to nasals long; lambdoidal crest well developed.

Comparisons.—From topotypes of *Thomomys bottae aureiventris*, *bonnevilliei* differs as follows: Size smaller, hind foot shorter. Color: Upper parts and sides lighter; underparts pale buff rather than "gold." Skull: Shorter and relatively wider; rostrum wider and heavier; zygomatic arches relatively wider and more massive, with greatest width posteriorly instead of anteriorly; interpterygoid space widely V-shaped rather than lyre-shaped; thickening at union of jugal and zygomatic process of maxilla less developed; anterior palatine foramina larger; nasals shorter and more markedly flared distally; zygomatic breadth relatively, and mastoidal breadth actually, wider; extension of premaxillae posterior to nasals greater; tympanic bullae more inflated ventrally; upper incisors wider.

From near topotypes of *Thomomys bottae centralis*, from 1 mile east of Garrison, Millard County, Utah, *bonnevilliei* differs as follows: Size smaller; hind foot and tail shorter. Color: Generally darker above and lighter below; top of head darker; postauricular patches smaller and lighter. Skull: Shorter and wider (zygomatic breadth expressed in percent of basilar length being, in males, 74.5 in *bonnevilliei* and 71.5 in *centralis*); interpterygoid space more widely V-shaped; interparietal smaller, and more triangular; nasals shorter and much more dilated distally, as well as more constricted proximally; lacrimal processes smaller and less globose at tips; temporal fossae larger; braincase and entire dorsal surface of skull more nearly flat; lambdoidal suture convex posteriorly as opposed to nearly straight; tympanic bullae more inflated ventrally.

Comparisons of *bonnevilliei* with the type and type series of *Thomomys bottae wahwahensis* show them to be of approximately the same size, but to differ as follows: Color: Slightly darker above and lighter below; postauricular patches smaller and lighter. Skull: Larger in every measurement taken, except breadth of rostrum which is smaller; skull not as flat; tympanic bullae more inflated ventrally; nasals and rostrum longer; extension of premaxillae posterior to nasals greater; interparietal smaller and more triangular; zygomatic arches more bowed out laterally; jugals heavier; interpterygoid space more widely V-shaped; upper incisors less massive.

The characters that distinguish *bonnevilliei* from *Thomomys bottae albicaudatus* are: Size smaller. Color: Markedly lighter throughout. Skull: Shorter and wider; mastoid and zygomatic breadths greater; rostrum narrower but shorter; angle between rostrum and zygomatic processes of maxillae less; interparietal smaller and more triangular; extension of premaxillae posterior to nasals greater; upper incisors shorter, narrower and more recurved.

T. b. bonnevilliei is indistinguishable in color from *Thomomys bottae convexus*, but differs from it in the following features: Size larger in nearly every measurement taken. Skull: Flattened dorsally as opposed to convex; zygomatic arches longer and weaker; jugals more nearly perpendicular; tympanic bullae larger; upper incisors longer; alveolar length of upper molar series the same, but molars narrower; rostrum longer but nasals shorter; extension of premaxillae posterior to nasals greater.

Topotypes of *bonnevillei* can be distinguished from those of both *Thomomys bottae tivius* and *stansburyi* by being larger in every measurement taken, by markedly lighter color throughout, and by ridged, massive, angular skulls rather than smooth, weak, nonangular skulls.

The races closest geographically to *bonnevillei* are *Thomomys bottae robustus* and *T. b. sevieri*. Compared with topotypes of *robustus*, *bonnevillei* differs in: Size larger. Color: Lighter throughout. Skull: Larger, although not as compact; zygomatic arches more widely spreading; jugals lighter; lacrimal processes not as prominent; zygomatic processes of maxillae not as robust; nasals more flared distally; extension of premaxillae posterior to nasals greater; alveolar length of upper molar series longer; molars larger; upper incisors longer, wider and darker in color; when placed ventral side down on a surface, the dorsal face of a skull of *robustus* is approximately parallel to the surface, whereas one of *bonnevillei* dips down in the occipital region.

T. b. sevieri can be easily distinguished from *bonnevillei* by being smaller in every measurement taken, darker in color, and by small, weak, smooth skulls as opposed to large, robust, ridged skulls.

Remarks.—Fish Springs, where *bonnevillei* occurs is a marshy area south of the barren, salt-desert country of western Utah. The source of water is springs at the base of the north end of the Fish Springs Mountains. Only the moist area supports pocket gophers. Specimens from Trout Creek, Juab County, twenty-five miles to the southwest are intergrades between *bonnevillei* and *aureiventris*, and are referred to the latter subspecies. The country between Fish Springs and Trout Creek in 1937 and 1940 lacked pocket gophers; it was of the playa and sand type. Probably *T. b. bonnevillei* was derived from *T. b. aureiventris*, a western mainland form of Pleistocene Lake Bonneville, through isolation and subsequent differentiation morphologically. The moist soils at Cane Springs, seven miles south of Fish Springs, had no pocket gophers when visited in 1940.

Specimens examined.—Total, 11, from the type locality.

***Thomomys bottae centralis* Hall**

Thomomys perpallidus centralis Hall, Univ. California Publ. Zoöl., 32:445, July 8, 1930.

Thomomys bottae centralis Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935; Hall and Johnson, Proc. Utah Acad. Sci. Arts and Letters, 15:121, 1938.

Type.—Male, adult, skin and skull, No. 41688, Museum of Vertebrate Zoölogy, University of California; 2-1/2 mi. E Baker (1-1/4 mi. W Nevada-Utah boundary on 39th parallel), 5,700 ft., White Pine County, Nevada; May 30, 1929; collected by E. Raymond Hall; original number 2683.

Range.—Extreme western Utah, in Millard, Beaver and Iron counties.

Diagnosis.—Size medium (see measurements); tail long; claws on front feet long. Color: Near Cinnamon Buff on upper parts, darker in middorsal region, grading to Pinkish Buff on underparts, more accentuated in pectoral and inguinal regions; nose, cheeks and postauricular patches grayish black; front and hind feet and distal half of tail white. Skull: Robust and moderately ridged; zygomatic breadth about the same for entire length of arches; jugals vertical posterior to middle; moderate thickening present at region of maxillo-jugal suture; interpterygoid space narrowly V-shaped; dorsal frontomaxillary sutures convex medially; lacrimal processes globose and well developed; nasals long and with distal denticulations; paroccipital processes well developed.

Comparisons.—Compared with topotypes of *Thomomys bottae albicaudatus*, *centralis* differs as follows: Size larger; tail longer; claws on front feet longer. Color: Lighter throughout, Cinnamon Buff as opposed to near (13"") Black. Skull: Basilar length and length of nasals greater; zygomatic breadth less; zygomatic arches thicker at region of maxillo-jugal sutures; interpterygoid space more broadly V-shaped; dorsal frontomaxillary sutures convex medially as opposed to straight; paroccipital processes more developed; zygomatic arches approximately the same width throughout as opposed to widest posteriorly.

For comparisons with *Thomomys bottae aureiventris* see account of that form.

T. b. centralis can be distinguished from *Thomomys bottae bonnevillei*, *robustus*, *sevieri* and *convexus* by larger size throughout and generally darker color (see accounts of those forms). From *Thomomys bottae stansburyi* and *tivius*, *centralis* differs in larger size throughout and lighter color (see accounts of those forms).

Remarks.—*Thomomys bottae centralis* has one of the most extensive ranges of any of the known races of *T. bottae*. The eastern limits extend into extreme western Utah. Specimens from Utah for the most part are intergrades between *centralis* and *aureiventris*, the race to the north. Some minor intergradation is also noted between *centralis* and *sevieri* and *bonnevillei*, the races to the east. Intergradation is the expected condition because the animals belonging to *centralis* are at the extremes of their range in this area. The greater affinities of these animals with *aureiventris* is to be expected because both *aureiventris* and *centralis* are forms of the western mainland of the Pleistocene Lake Bonneville; while the races to the east, although closest geographically, were isolated from the gophers of the western mainland during prehistoric times by this lake. They are still isolated and enough time has elapsed so that only vestiges of morphological intergradation exist between *centralis* and these eastern forms. Two specimens from Cedar City, Iron County, are intergrades between *Thomomys bottae wahwahensis*, *centralis* and *planirostris*. Their skulls are slightly convex as in *planirostris*, and the rostrum is short and wide as in *wahwahensis*. In shape of the zygomatic arches, length of the nasals, and color, they resemble *centralis* to which they are here referred.

Specimens examined.—Total, 49, distributed as follows: *Millard County*: 1 mi. SE Gandy, 5,000 ft., 15 (M. V. Z.); White Valley (Tule Spring), 60 mi. W Delta, 4, (3 in R. W. Fautin Vertebrate Collection); Robison Ranch, 5,300 ft., (on Hendry Creek) Simonsons Ranch, 4,596 ft., 2 (M. V. Z.); 1 mi. E Garrison, 5,000 ft., 21; 5 mi. S Garrison, 5,400 ft., 5 (M. V. Z.). *Iron County*: Cedar City, 2 (M. V. Z.).

***Thomomys bottae sevieri* new subspecies**

Type.—Female, adult, skin and skull, No. 2530, Museum of Zoölogy, University of Utah; Swasey Spring, House Mountains, 6,500 ft., Millard County, Utah; May 16, 1938; collected by S. D. Durrant; original number 1380.

Range.—Known only from the type locality.

Diagnosis.—Size medium (see measurements); claws on front feet short and weak; ears short; tail relatively long. Color: Upper parts Pinkish Buff, grading over sides to Pale Pinkish Buff on underparts; nose, top of head, chin and cheeks grayish black; postauricular patches small and grayish black; front and hind feet and distal two-thirds of tail white. Skull: Small, weak and smooth; rostrum narrow; nasals narrow, not markedly flared distally; zygomatic arches weak, not angular, and of "graceful" contour; lacrimal processes small; characteristic dorsal depression present in region of sagitto-coronal suture; mastoid and zygomatic breadths narrow; occiput narrow and high; braincase well inflated; paroccipital processes small and smooth; interpterygoid space narrowly V-shaped; tympanic bullae small, but well inflated ventrally; alveolar length of upper molar series short; molars small; upper incisors short, but narrow.

Comparisons.—From topotypes of *Thomomys bottae aureiventris*, *sevieri* differs as follows: Size smaller. Color: Lighter throughout, no "gold" on underparts. Skull: Much smaller in every measurement taken, less massive and not angular; zygomatic arches weaker and widest posteriorly rather than anteriorly; union of jugal and zygomatic process of maxilla not greatly thickened; interpterygoid space narrowly V-shaped rather than lyre-shaped; pterygoid hamulae shorter and weaker; tympanic bullae smaller, but markedly more inflated ventrally; dentition smaller and weaker.

From near topotypes of *Thomomys bottae centralis*, *sevieri* can be distinguished by the following features: Size markedly smaller. Color: Lighter throughout. Skull: Markedly smaller in every measurement taken, weaker and smoother; zygomatic arches weaker, less angular and more "graceful"; rostrum shorter, but narrower; lacrimal processes smaller; tympanic bullae smaller, but more inflated ventrally, being triangular in shape as opposed to ovate and with anteromedial margin decidedly pointed; pterygoid hamulae smaller and weaker; dentition smaller and weaker.

T. b. sevieri can readily be distinguished from *Thomomys bottae albicaudatus* by the following features: Size smaller in every measurement taken. Color: Markedly lighter throughout. Skull: Smaller, and weaker; rostrum shorter and narrower; ascending processes of premaxillae narrower; extension of premaxillae posterior to nasals shorter; posterior tongues of premaxillae narrower; dentition much lighter.

Comparisons of *sevieri* with topotypes of *Thomomys bottae wahwahensis* show them to be of approximately the same size, but to differ as follows: Hind foot longer; ear shorter. Color: Slightly darker. Skull: Smaller, weaker, less ridged; zygomatic breadth less; zygomatic arches markedly less angular; mastoid breadth less; rostrum much

longer and narrower, not as blunt nor flattened; tympanic bullae much larger and more inflated ventrally; braincase vaulted as opposed to flattened.

From topotypes of *Thomomys bottae bonnevilliei*, *sevieri* differs in: Size smaller throughout. Skull: Smaller in every measurement taken, weaker, smoother and less angular; dentition smaller and weaker.

Topotypes of *sevieri* are easily distinguished from those of *Thomomys bottae robustus* by smaller size, and smaller, markedly weaker skull which is less angular and ridged.

Among named races of *Thomomys bottae*, *sevieri* is closest geographically to *convexus*, but differs from it as follows: Size larger; hind foot longer. Skull: Smaller in every measurement taken; nasals shorter and not so flaring distally; rostrum weaker, narrower and not so depressed; zygomatic arches markedly weaker and less angular; lacrimal processes smaller; supraoccipital narrower and higher; paroccipital processes weaker; tympanic bullae smaller; dentition markedly weaker.

Topotypical specimens of *sevieri* can be readily distinguished from those of *Thomomys bottae tivius* by Pinkish Buff instead of Mummy Brown on upper parts. Tympanic bullae larger and markedly more inflated; nasals longer; zygomatic and mastoidal breadths greater; rostrum longer and more depressed; upper incisors longer and wider; molariform teeth smaller. The skulls of *sevieri* resemble those of *tivius* more closely than those of any other subspecies.

Remarks.—The House Mountains in western Millard County are surrounded by desertlike terrain that is seemingly unsuited to pocket gophers. In these mountains, gophers were sought in vain at several localities, including Antelope Springs which superficially appeared suitable for the animals. Pocket gophers were found only at the type locality, Swasey Spring, which is well above the highest level of the Pleistocene Lake Bonneville. *T. b. sevieri*, like *T. b. minimus* on Stansbury Island, Great Salt Lake, appears to remain only on land that was an island when Lake Bonneville was at its highest level.

Specimens examined.—Total, 10, from the type locality.

***Thomomys bottae convexus* Durrant**

Thomomys bottae convexus Durrant, Proc. Biol. Soc. Washington, 52:159, October 11, 1939.

Type.—Male, adult, skin and skull, No. 2482, Museum of Zoölogy, University of Utah; E side Clear Lake, 4,600 ft., Millard County, Utah; May 20, 1938; collected by S. D. Durrant; original number 1401.

Range.—Westcentral Utah in Delta Valley.

Diagnosis.—Size medium (see measurements). Color: Upper parts and sides Pinkish Buff, purest on sides; underparts Pale Pinkish Cinnamon; inguinal and pectoral regions Pale Pinkish Buff; nearly all specimens have white on perineal region; nose grayish black; front feet, hind feet and distal third to half of tail white; postauricular patches black. Skull: Braincase moderately convex on dorsal surface; rostrum strongly depressed, giving the entire dorsal surface of the skull a "rocker-shape"; zygomatic arches heavy, short and widely spreading, widest posteriorly; upper incisors recurved, short and wide; molariform teeth large; alveolar length of upper molar series long; palatal pits deep; tympanic bullae moderately inflated ventrally; mastoidal breadth actually as well as relatively wide.

Comparisons.—Compared with topotypes of *Thomomys bottae wahwahensis*, *convexus* is of approximately the same color, but differs as follows: Size smaller; tail, hind foot, and ear shorter. Skull: Rostrum longer, narrower and more depressed; skull convex rather than flat; nasals longer, and convex rather than flat; tympanic bullae larger; zygomatic arches shorter and more massive; molariform teeth larger.

From topotypes of *Thomomys bottae centralis*, *convexus* differs in: Size smaller; tail and hind foot shorter. Color: Uniformly lighter, more white in perineal region. Skull: Smaller, more convex; rostrum shorter, wider and more depressed; zygomatic arches shorter and heavier; mastoidal breadth actually, as well as relatively wider; tympanic bullae more inflated ventrally; upper incisors shorter and wider.

Comparatively, topotypes of *convexus* can be distinguished from those of *Thomomys bottae aureiventris* by: Size smaller; tail and hind foot shorter. Color: Darker on upper parts; no "gold" on underparts. Skull: Smaller and more

nearly flat; rostrum shorter and more depressed; zygomatic arches shorter, heavier and widest posteriorly rather than anteriorly; interpterygoid space V-shaped as opposed to lyre-shaped; upper incisors shorter, narrower and more recurved.

Topotypical specimens of *convexus* differ from those of *Thomomys bottae nesophilus* as follows: Size smaller; tail and hind foot shorter. Color: Uniformly lighter throughout, Cinnamon Buff as opposed to Pinkish Buff. Skull: Smaller; rostrum heavier, shorter and more depressed; zygomatic arches shorter, heavier and not so widely spreading; no widening of supraoccipital as in *nesophilus*; upper incisors shorter and more recurved.

When compared with topotypes of *Thomomys bottae albicaudatus*, *convexus* shows the following differences: Size smaller; tail and hind foot shorter. Color: Markedly lighter throughout. Skull: Smaller, more convex and compact; rostrum shorter, heavier, more depressed and compact; zygomatic arches shorter and more robust; upper incisors shorter and more recurved.

Thomomys bottae tivius is the race closest geographically to *convexus*. From it, *convexus* can be readily distinguished by: Size larger; tail shorter; hind foot longer. Color: Markedly lighter throughout. Skull: Much heavier and more compact, weights of skulls of males and females of the two subspecies being 2.4 grs., 1.6; 1.6, 1.2, respectively; rostrum heavier, wider and more depressed; zygomatic arches shorter, and more massive; upper incisors shorter, wider and more recurved; molariform teeth larger.

For comparisons with *Thomomys bottae lenis*, *contractus*, *sevieri*, *bonnevillei*, and *robustus* see accounts of those forms.

Remarks.—*T. b. convexus* is limited to the area around Clear Lake in Millard County. This lake is surrounded by areas of loose, shifting sand and flat areas of barren alkali. The lake is fed by springs which flow from lava outcroppings on its eastern side. As far as discernible, the only area populated by pocket gophers (1938) was that adjacent to the lake where vegetation had trapped the sand. The factor which limits the extension of range of this subspecies probably is plant food. Also, the soil is mechanically poor for burrowing, since it caves in easily and burrows were found only in the sand where salt grass (*Distichlis stricta*) had trapped and stabilized it. Burrows were found from the edge of the water back as far as this grass persisted.

Specimens examined.—Total, 17, from the type locality.

***Thomomys bottae tivius* Durrant**

Thomomys bottae tivius Durrant, Bull. Univ. Utah, 28 (No. 4):5, August 18, 1937.

Type.—Female, adult, skin and skull, No. 1827, Museum of Zoölogy, University of Utah; Oak Creek Canyon, 6 mi. E Oak City, 6,000 ft., Millard County, Utah; September 14, 1936; collected by S. D. Durrant; original number 1100.

Range.—Limited to the Cañon Mountains, Millard County.

Diagnosis.—Size small (see measurements). Color: Upper parts Mummy Brown, grading through Cinnamon on the sides to Pale Cinnamon on the underparts; cheeks Cinnamon; postauricular patches black; distal third to half of tail white. Skull: Small, weak; zygomatic arches weak, not widely spreading, widest posteriorly; tympanic bullae large; interpterygoid space V-shaped; nasals short, usually simple distally, but with some denticulations in some specimens; palatal pits deep; palate narrow; paroccipital processes small; incisors, both upper and lower, narrow; molariform teeth small.

Comparisons.—Topotypes of *tivius* differ from those of *Thomomys bottae albicaudatus* as follows: Size markedly smaller in every measurement taken. Color: Lighter, Mummy Brown as opposed to near (13"n) Black. Skull: Smaller, slenderer and weaker; zygomatic arches weak and not widely spreading as opposed to massive and wide spreading; nasals and rostrum narrower and shorter; extension of premaxillae posterior to nasals shorter; tympanic bullae smaller; molariform teeth smaller.

For comparisons with *Thomomys bottae stansburyi* and *T. b. contractus* see accounts of those forms.

The four subspecies *tivius*, *albicaudatus*, *stansburyi*, and *contractus* are the darkest in color of all the *Thomomys bottae* occurring within the state.

Remarks.—This small, dark subspecies is limited to the Cañon Mountains in eastern Millard County. Apparently it is a mountain derivative of *Thomomys bottae contractus* which occurs in the valleys to the east and west of these mountains. Intergradation is noted with animals from the valleys on either side. For further comments on distributional problems of this type see remarks under *Thomomys bottae stansburyi*.

Specimens examined.—Total, 12, from the type locality.

***Thomomys bottae contractus* new subspecies**

Thomomys perpallidus albicaudatus Hall, Univ. California Publ. Zoöl., 37:3, April 10, 1931.

Thomomys bottae albicaudatus Durrant. Bull. Univ. Utah, 28 (No. 4):4, August 18, 1937.

Type.—Male, adult, skin and skull, No. 1851, Museum of Zoölogy, University of Utah; Scipio, 5,315 ft., Millard County, Utah; September 17, 1936; collected by S. D. Durrant; original number 1125.

Range.—Extreme eastern Millard and Beaver counties, Utah.

Diagnosis.—Size medium (see measurements). Color: Upper parts Cinnamon Buff, mixed with black giving a color of Dresden Brown; sides between Cinnamon Buff and Pinkish Buff; underparts Pinkish Buff, purest on inguinal and pectoral regions; postauricular patches medium in size and black; ears covered with black hairs; nose, chin, cheeks and top of head dusky; front feet, hind feet and distal third to half of tail white; proximal part of tail covered all around with buff-colored hairs. Skull: Long, slender, moderately ridged and convex transversally at proximal ends of nasals; nasals long; rostrum long and narrow; posterior ends of nasals truncate or shallowly emarginate; ascending processes of premaxillae slender; extension of premaxillae posterior to nasals long; zygomatic arches neither robust nor widely spreading; interparietal subquadrangular; supraoccipital extending horizontally well behind lambdoidal suture instead of dropping off abruptly to the foramen magnum; interpterygoid space moderately V-shaped in some specimens, but somewhat lyre-shaped in others; tympanic bullae large and truncate anteriorly; upper incisors long and narrow; molariform teeth small and light.

Comparisons.—Compared with topotypes of *Thomomys bottae albicaudatus*, *contractus* differs as follows: Tail longer. Color: Lighter throughout. Skull: Slenderer, less ridged and angular; rostrum narrower; zygomatic and mastoidal breadths less; ascending processes of premaxillae narrower; posterior tongues of premaxillae narrower; posterior ends of nasals less truncate; zygomatic arches weaker, less angular, and less widely spreading posteriorly; interparietal larger; paroccipital processes weaker; interpterygoid space not as widely V-shaped; upper incisors longer and narrower; molariform teeth smaller.

Topotypes of *contractus* can be distinguished from those of *Thomomys bottae convexus* by the following: Size larger, tail longer; hind foot larger. Color: Darker throughout. Skull: Longer, narrower, and not as massive; top of skull moderately, as opposed to strongly, convex; nasals arched rather than straight; zygomatic arches neither as widely spreading, angular nor massive; space enclosed within zygomatic arches longer; interparietal larger; interpterygoid space more narrowly V-shaped; upper incisors longer and narrower; molariform teeth much lighter.

Comparisons of topotypes of *contractus* with near topotypes of *Thomomys bottae centralis* show them to be approximately the same size, but to differ as follows: Color: Darker throughout. Skull: Shorter and slenderer; rostrum narrower; region between posterior tongues of premaxillae narrower and more convex transversally; nasals more truncate; zygomatic breadth less, but arches relatively more widely spreading posteriorly; interparietal larger; interpterygoid space generally narrower; upper incisors longer and narrower; molariform teeth smaller.

Topotypes of *contractus* differ from those of *Thomomys bottae aureiventris* as follows: Size smaller; tail longer; hind foot shorter. Color: Darker throughout. Skull: Shorter but slenderer; rostrum narrower; nasals shorter but slenderer, and more truncate posteriorly; extension of premaxillae posterior to nasals longer; zygomatic arches weaker and less angular; zygomatic processes of maxillae weaker and with no marked thickenings at union of maxilla and jugals; interparietal larger; interpterygoid space more generally V-shaped; upper incisors longer and narrower; molariform teeth smaller.

Compared with topotypes of *Thomomys bottae planirostris*, *contractus* differs in: Size smaller throughout. Color: Darker, more black and less Cinnamon in pelage. Skull: Smaller in every measurement taken; rostrum narrower;

nasals arched instead of flat; zygomatic arches neither angular, massive nor widely spreading; upper incisors narrower; molariform teeth markedly smaller and weaker.

Topotypes of *contractus* differ from those of *Thomomys bottae levidensis* in larger size, darker color and longer, slenderer skulls.

Among named races of *T. bottae*, *contractus* is closest morphologically to *tivius*. It differs from it as follows: Size larger throughout. Color: Lighter throughout. Skull: The same general shape and proportions, but larger in every measurement taken; rostrum longer and narrower; extension of premaxillae posterior to nasals longer; posterior tongues of premaxillae narrower.

Remarks.—Fifteen animals from Oak City are intergrades between *contractus* and *tivius*. Intergradation with *lenis* is also shown in some specimens by the widely spreading zygomatic arches. In the majority of characters including the diagnostic long, slender, narrow rostrum they are more like *contractus* to which they are here referred.

Nine animals from Beaver were considered by Hall (1931:3) and Durrant (1937:4) to be intergrades between *Thomomys bottae albicaudatus* and *Thomomys bottae centralis*. Restudy of these specimens in the light of additional material now shows them to be intergrades between *T. b. centralis*, *T. b. planirostris* and *T. b. contractus*. The majority of these animals are intermediate in color between *centralis* and *contractus*, but a few have the reddish cast of *planirostris*. The shape of the nasals is characteristic of *planirostris*, while the zygomatic arches are as in *centralis*. In the remainder of the diagnostic characters they are like *contractus* to which they are here referred.

Strong affinities exist between *albicaudatus*, *tivius* and *contractus*. All three of these races probably stemmed from a dark form which formerly inhabited the eastern mainland of the Pleistocene Lake Bonneville. At present, *tivius* is isolated on the Cañon Mountains in eastern Millard County, while the range of *albicaudatus* and *contractus* have been separated by that of *lenis*. *T. b. lenis* has the majority of its affinities with *aureiventris* which is an inhabitant of the western mainland of this ancient lake. An understanding of the history of the Sevier River Valley will probably clarify this distribution of pocket gophers.

Specimens examined.—Total, 39, distributed as follows: *Millard County*: Oak City, 5,000 ft., 15; Scipio, 5,315 ft., 15. *Beaver County*: Beaver, 6,000 ft., 9 (M. V. Z.).

***Thomomys bottae lenis* Goldman**

Thomomys townsendii lenis Goldman, Proc. Biol. Soc. Washington, 55:75, June 25, 1942.

Thomomys perpallidus aureus Moore, Journ. Mamm., 10:259; November 11, 1931.

Type.—Male, adult, skin and skull, No. 264805, U. S. National Museum (Biological Surveys Collection); Richfield, 5,308 ft., Sevier County, Utah; March 11, 1928; collected by A. W. Moore; X-catalogue number 28835 (after Goldman, type not seen).

Range.—Sevier River Valley from Piute County north to southwestern Juab and northeastern Millard counties, Utah.

Diagnosis.—Size large (see measurements). Color: Upper parts Cinnamon Buff mixed with black in middorsal region; sides, flanks, forearms, thighs and underparts Pinkish Buff; inguinal region, front feet, hind feet, underpart of tail and end of tail white; postauricular patches small and dusky; chin, cheeks, nose and top of head dusky. Skull: Largest of Utah gophers, massive and angular; nasals long and denticulate distally; rostrum long and relatively narrow; zygomatic arches widely spreading and heavy throughout; jugals nearly vertical; zygomatic processes of maxillae heavy and flaring out abruptly from base of rostrum; union of zygomatic process of maxilla and jugal greatly thickened; extension of premaxillae posterior to nasals long; posterior tongues of premaxillae relatively narrow; lacrimal processes small; pterygoid hamulae long; interpterygoid space moderately V-shaped, tending to be somewhat lyre-shaped in some specimens; tympanic bullae somewhat flattened, only moderately inflated ventrally; upper incisors long and narrow; molariform teeth actually large, but relatively small.

Comparisons.—Topotypes of *lenis* can be distinguished from those of *Thomomys bottae tivius*, *convexus*, *contractus*, *albicaudatus*, *levidensis*, *centralis* and *aureiventris* by the following markedly greater average measurements of males: Total length, 250 mm.; length of nasals, 15.5; zygomatic breadth, 28.3; mastoid breadth,

22.5; and length of rostrum, 18.3. Other distinguishing characters are: Zygomatic arches more widely spreading; length of zygomatic processes of maxillae greater; and relatively longer, narrower rostrum.

Remarks.—Twenty-one animals obtained from Lynndyl, Millard County, are all intergrades between *lenis* and *aureiventris*. They are like *aureiventris* in the shape of the zygomatic arches, and in the bowing of the parietal crests. Slight intergradation with *centralis* is indicated by color and the shape of the nasals. The transverse arching of the posterior part of the rostrum is indicative of some relationship with *contractus*. In six other characters studied they most closely approach *lenis* to which they are here referred.

Large size is the distinctive feature of *Thomomys bottae lenis*. The skulls are the largest of any species or subspecies of *Thomomys* found in Utah. In total length, however, these animals are no longer than the extremes found in other named races. When Goldman (1942:75) described this race as new, he referred it to the species *Thomomys townsendii*, but remarked that the animal from Richfield was different enough from any other form then named to merit probably full specific status. I know of no character other than size to separate *Thomomys townsendii* from *Thomomys bottae*, and since intergradation has been shown to exist between these alleged *townsendii* from Richfield and animals from extreme western Utah known to belong to the species *bottae*, *lenis* is here arranged as a subspecies of *Thomomys bottae* which name has priority over *Geomys townsendii*.

The range here ascribed to this race is the Sevier River Valley from Piute County as far downstream as the town of Lynndyl which is near the eastern mainland of Pleistocene Lake Bonneville. The Sevier River continues farther out into Delta Valley ultimately to empty into Sevier Lake, which at present is adjacent to the area that formerly constituted the western mainland of the aforementioned ancient lake. This watercourse may have provided a migration route in ancient times, during the fluctuations of Lake Bonneville, whereby the animals formerly of the western mainland were able to come far eastward. The animals from Lynndyl which are intergrades between *lenis*, an eastern mainland form, and *centralis* and *aureiventris* which are western mainland forms of Lake Bonneville lend support to this hypothesis.

Specimens examined.—Total, 26, distributed as follows: *Millard County*: Lynndyl, 4,796 ft., 21. *Juab County*: U. B. (= Yuba) Dam, 5,000 ft., 1. *Sevier County*: Salina, 4,575 ft., 1; Richfield, 5,308 ft., 3. (U. S. N. M.).

***Thomomys bottae levidensis* Goldman**

Thomomys bottae levidensis Goldman, Proc. Biol. Soc. Washington, 55:76, June 25, 1942.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—Male, adult, skin and skull, No. 191962, U. S. National Museum (Merriam Collection); Manti, 5,500 ft., Sanpete County, Utah; December 6, 1888; collected by Vernon Bailey; original number 427 (after Goldman, type not seen).

Range.—San Pitch River Valley, Sanpete County, Utah.

Diagnosis.—Size small (see measurements). Color: Upper parts and sides Cinnamon Buff, finely mixed with black along median line of back; underparts Pinkish Buff; nose, cheeks and chin grayish black; postauricular patches fairly large and grayish black; front and hind feet white (examples from type series badly stained); tail light buff but apparently white distally (the color of these specimens has apparently changed with age). Skull: Small, fairly robust; basilar length short; zygomatic arches weak, but widely spreading; tympanic bullae small; nasals short and simple distally; ventral margin of jugals convex dorsally; extension of premaxillae posterior to nasals relatively as well as actually long; posterior tongues of premaxillae relatively wide.

Comparisons.—Topotypes of *levidensis* differ from those of *Thomomys bottae absonus* as follows: Size smaller. Color: Lighter throughout. Skull: Shorter, weaker and less ridged and angular, but relatively wider.

Compared with topotypes of *Thomomys bottae albicaudatus*, *levidensis* differs as follows: Size smaller in every measurement taken. Color: Markedly lighter throughout. Skull: Smaller in every measurement taken; width relatively greater; skull smooth, weak and nonangular as opposed to ridged, robust and angular.

For comparisons with *Thomomys bottae lenis* and *contractus* see accounts of those forms.

Remarks.—The range here ascribed to *levidensis* is the San Pitch River Valley, which gradually merges southward into the Sevier River Valley. The latter valley in this area is inhabited by pocket gophers that belong to another subspecies, *lenis*. Nephi Valley to the west of San Pitch River Valley is inhabited by animals belonging to the subspecies *albicaudatus*. No known specimens show intergradation between *lenis* and *levidensis*, but intergradation between *lenis* and *albicaudatus* is noted in the Nephi Valley animals (see account of *albicaudatus*). Superficially *levidensis* resembles *absonus* in size and color, but the skulls closely resemble those of *albicaudatus*, except for size in which they are smaller in all measurements. *T. b. albicaudatus* is the most variable subspecies of *T. bottae* occurring in Utah, and additional material from the Sevier River Valley between San Pitch River Valley and Nephi Valley may show *levidensis* to be only a local variant of the highly variable subspecies, *albicaudatus*.

Specimens examined.—Total, 6, from the type locality.

***Thomomys bottae osgoodi* Goldman**

Thomomys perpallidus osgoodi Goldman, Journ. Washington Acad. Sci., 21:424, October 19, 1931.

Thomomys bottae osgoodi Goldman, Proc. Biol. Soc. Washington, 48:156; October 31, 1935.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—Male, adult, skin and skull, No. 158530, U. S. National Museum (Biological Surveys Collection); Hanksville, Wayne County, Utah; October 20, 1908; collected by W. H. Osgood; original number 3701 (after Goldman, type not seen).

Range.—Eastern Utah in the valleys of the drainage of the San Rafael, Dirty Devil and Price rivers.

Diagnosis.—Size medium (see measurements). Color: Upper parts near (e) Pale Ochraceous Buff, definitely yellow in appearance; sides Pale Ochraceous Buff; entire underparts white, with a wash of Light Buff in the pectoral and inguinal regions; top of head, nose, cheeks, and chin dusky; postauricular patches grayish black; front feet, hind feet and distal part of tail white. Skull: Fairly robust but narrow; zygomatic arches concave medially in mid-jugal region; skull moderately convex dorsally, due to swelling in region of base of rostrum; lambdoidal suture situated well ahead of posterior margin of skull, with supraoccipital forming a side shelf at posterior part of skull; interpterygoid space narrowly V-shaped; tympanic bullae well inflated ventrally; basioccipital short; nasals rounded posteriorly; molariform teeth large.

Comparisons.—Topotypes of *osgoodi* differ from those of *Thomomys bottae absonus* as follows: Size generally smaller. Color: Lighter throughout, more yellowish in appearance as opposed to buffy. Skull: Smaller in all measurements, except length of nasals, mastoid breadth, and alveolar length of upper molar series which are larger; rostrum shorter but relatively wider; zygomatic arches more robust and concave medially; palate wider; supraoccipital more bulging posteriorly; tympanic bullae more inflated ventrally; molariform teeth larger.

For comparisons with *Thomomys bottae aureus* and *T. b. dissimilis* see accounts of those forms.

Remarks.—The animals here referred to *osgoodi* are remarkably uniform in color, but vary widely in cranial details. Specimens from Carbon County are not typical and when more material becomes available it may prove that these animals from the northern part of the range of *osgoodi* will merit separation and naming. The specimens from Emery County are not typical but resemble *osgoodi* more than do the animals from Carbon County.

The range here ascribed to *osgoodi* is in that part of the eastern Utah desert that is bounded on the east by the Green and Colorado rivers, on the west by the high mountains of central Utah, on the north by the Book Cliffs and on the south by the Dirty Devil River. This area is an uninviting wasteland in which there are relatively few roads and little water. In addition, it is greatly cut up by washes and gullies which contain water only during a few weeks of the year. The continuation of this area of wasteland southward beyond the Dirty Devil River is inhabited by pocket gophers belonging to the subspecies *absonus*. If specimens were available they would undoubtedly show intergradation to exist between *osgoodi* and *absonus*.

Specimens examined.—Total, 14, distributed as follows: *Carbon County*: 1-2 mi. N Spring Glen, 6,150 ft., 2; Spring Glen, 6,200 ft., 2; 2 mi. E Spring Glen, 6,200 ft., 1. *Emery County*: Price River, 2 mi. SE Woodside, 4,600 ft., 2 (C. M.); Green River, 4,080 ft., 5 (M. V. Z.). *Wayne County*: Hanksville, 2 (U. S. N. M.).

Thomomys bottae howelli Goldman

Thomomys bottae howelli Goldman, Journ. Washington Acad. Sci., 26:116, March 15, 1936.

Type.—Female, adult, skin and skull, No. 25684, U. S. National Museum (Biological Surveys Collection); Grand Junction, 4,600 ft., Mesa County, Colorado; November 7, 1895; collected by A. H. Howell; original number 493 (after Goldman, type not seen).

Range.—In the valleys of eastern Utah, east of the Green River and north of the Colorado River.

Diagnosis and Comparisons.—Inasmuch as there is but one specimen, the holotype known, and as it was impossible to study it, the following diagnoses and comparisons are from Goldman, (1936:116).

"*General characters*.—A rather large, pallid subspecies with a broad, flattened cranium. Similar to the palest specimens of *Thomomys bottae aureus* of the San Juan River Valley, southeastern Utah, in color, but underparts more thinly overlaid with buffy white, and cranial characters, especially the broad, flat braincase, distinctive. Approaching *Thomomys bottae osgoodi* of the Fremont River Valley, Utah, in color, but much larger and skull widely different.

"*Color*.—Type (winter pelage): Upper parts in general between tulle buff and pale olive buff (Ridgway 1912), somewhat darkened on head by a mixture of cinnamon buff and brown; a few inconspicuous dusky-tipped hairs along median line of back; muzzle dusky; ears and postauricular spots deep, contrasting black; underparts thinly overlaid with buffy white, the hairs becoming pure white to roots on inguinal region; thighs pure white to roots all around; feet white; tail buffy whitish, slightly paler below than above.

"*Skull*.—Similar in general to that of *T. b. aureus*, but braincase conspicuously broader and flatter; zygomatics more widely spreading; nasals shorter; premaxillae more attenuate posteriorly; interparietal larger; auditory bullae more rounded and fully inflated anteriorly; incisors short, as in *aureus*, but less strongly recurved. Compared with that of *T. b. osgoodi* the skull is much larger, with flatter braincase, shorter nasals, and posteriorly narrower premaxillae."

Remarks.—Six specimens, in the Carnegie Museum from 10 miles north of Moab, Grand County, Utah, were available for this study. They are not typical of *howelli* as it is diagnosed by Goldman (*loc. cit.*). They appear to be intergrades between *howelli* and *osgoodi* in cranial characters, but more closely resemble *howelli*, particularly in the flat, widened, low braincase. In color, some specimens seem to intergrade toward *aureus*.

The range ascribed to this form in Utah appears to be one of the most natural ones within the state since it is bounded by the Green and Colorado rivers which have formed deep rocky gorges in this region.

Specimens examined.—Total, 6, as follows: *Grand County*: 10 mi. N Moab, 6 (C. M.).

Thomomys bottae wahwahensis Durrant

Thomomys bottae wahwahensis Durrant, Bull. Univ. Utah, 28 (No. 4):4, August 18, 1937.

Type.—Male, adult, skin and skull, No. 1750, Museum of Zoology, University of Utah, Wah Wah Springs, 30 mi. W Milford, 6,500 ft., Beaver County, Utah; July 22, 1936; collected by S. D. Durrant; original number 989.

Range.—Westcentral Utah, in Wah Wah Mountains, and Pine Valley to the west of these mountains.

Diagnosis.—Size medium (see measurements). Color: Upper parts Pinkish Buff; underparts Pale Pinkish Buff with considerable admixture of gray; inguinal and pectoral regions Pale Pinkish Buff; nose and cheeks grayish black; postauricular patches small and black; front feet, hind feet and distal one-third to one-half of tail white. Skull: Flat dorsoventrally; rostrum short and wide; premaxillae broad and heavy; nasals short and straight, with no arching as viewed laterally; tympanic bullae small; space enclosed within zygomatic arches short antero-posteriorly; alveolar length of upper molar series short; molariform teeth small.

Comparisons.—From topotypes of *Thomomys bottae centralis*, *wahwahensis* differs as follows: Size smaller in every measurement taken. Color: Lighter, Pinkish Buff as opposed to Cinnamon Buff. Skull: Rostrum wider, shorter and more nearly flat; nasals straight as opposed to moderately convex; tympanic bullae smaller and less

inflated ventrally; zygomatic arches more widely spreading and angular; molariform teeth smaller; extension of premaxillae posterior to nasals less.

From topotypes of *Thomomys bottae albicaudatus*, *wahwahensis* differs as follows: Hind foot shorter. Color: Lighter throughout, Pinkish Buff as opposed to (13'''n) Black. Skull: Smaller and more nearly flat; rostrum shorter, wider and more nearly flat; nasals straight as opposed to convex; zygomatic breadth less but mastoid breadth greater; tympanic bullae smaller, and less inflated ventrally; extension of premaxillae posterior to nasals less; molariform teeth smaller.

From topotypes of *Thomomys bottae aureiventris*, *wahwahensis* differs in the following features: Size smaller; hind foot shorter. Color: Lighter throughout, no "gold" on underparts. Skull: Smaller in nearly every measurement taken; rostrum shorter and relatively wider; zygomatic arches more angular and relatively more widely spreading; nasals shorter and more nearly flat; thickening at union of jugal and zygomatic process of maxilla less; interpterygoid space V-shaped as opposed to lyre-shaped; tympanic bullae much smaller, and less inflated ventrally; molariform teeth much smaller.

Topotypes of *wahwahensis* can be easily distinguished from those of *Thomomys bottae tivirus* by their markedly larger size in every measurement taken, lighter color, and larger, more robust and more nearly flat skull.

For comparisons of *wahwahensis* with *Thomomys bottae sevieri*, *robustus*, *bonnevillei* and *convexus* see comparisons under those forms.

Among the named races of *Thomomys bottae*, *wahwahensis* definitely has its affinities with *planirostris* from Zion National Park. Both possess flat skulls with wide, short rostra. It differs from the latter in: Size smaller in every measurement taken. Color: Lighter throughout. Skulls: Nasals and rostrum shorter and more nearly flat; tympanic bullae markedly smaller; alveolar length of upper molar series shorter; molariform teeth markedly smaller and weaker.

Remarks.—Wah Wah Springs, the type locality of *wahwahensis*, are on the summit of a low pass in the Wah Wah Mountains in the desert of west central Utah. The surrounding valleys, for many miles, as far as my investigations show, are not inhabited by pocket gophers, except the Desert Range Experiment Station of the United States Forest Service in Pine Valley to the west of these mountains. There, pocket gophers were obtained which are intergrades between *centralis* and *wahwahensis*. In five out of seven characters investigated these gophers resemble *wahwahensis*, to which they are here referred. Study of the topography reveals the probable means by which the animals reached this valley. The long axis of the Wah Wah Mountains is north and south, but a westward arm forms the northern boundary of Pine Valley. Around springs in this westward projecting arm workings of pocket gophers were found. With the development of water at the Desert Range Experiment Station, and subsequent improvement of forage, these animals probably came down into the valley from the springs to the north.

The terrain between the Desert Range Experiment Station in Pine Valley and Snake Creek (where *centralis* occurs) to the west is not inhabited by pocket gophers at present. This area, however, forms part of the southwest mainland of Pleistocene Lake Bonneville, which mainland in times past was probably suitable for pocket gophers. Since the close of the Pleistocene, aridity has rendered most of it unfit for pocket gophers, and they remain only in isolated areas where suitable environments still persist.

Specimens examined.—Total, 18, distributed as follows: *Millard County*: Desert Range Experiment Station, United States Forest Service, Sec. 9, T. 25 S, R. 17 W, Salt Lake Base Meridian, 6. *Beaver County*: Wah Wah Springs, Wah Wah Mountains, 30 mi. W Milford, 6,500 ft., 12 (2, M. V. Z.).

***Thomomys bottae dissimilis* Goldman**

Thomomys perpallidus dissimilis Goldman, Journ. Washington Acad. Sci., 21:425, October 19, 1931.

Thomomys bottae dissimilis Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—Female, adult, skin and skull, No. 158526, U. S. National Museum (Biological Surveys Collection); E slope Mount Ellen, Henry Mountains, 8,000 ft., Garfield County, Utah; October 15, 1908; collected by W. H.

Osgood; original number 3677 (after Goldman, type not seen).

Range.—Known only from the type locality.

Diagnosis.—Size small (see measurements). Color: Upper parts Light Buff, grading over sides to nearly white on underparts; underparts lightly washed with Pale Buff, more marked in inguinal and pectoral regions; postauricular patches grayish black; nose, chin, cheeks and top of head dusky; front feet, hind feet and distal half of tail white. Skull: Small and weak; zygomatic arches long, but lying close to skull, giving it a slender appearance; supraoccipital markedly projecting posteriorly from lambdoidal suture; rostrum relatively long and narrow; nasals long; tympanic bullae well inflated ventrally, with a median ventral ridge; pterygoid hamulae weak; interpterygoid space narrowly V-shaped; upper incisors short and light in color; molariform teeth relatively large.

Comparisons.—Comparison of one topotype of *dissimilis* with topotypes of *Thomomys bottae aureus* shows it to differ as follows: Size smaller throughout. Color: Lighter dorsally and on sides, pale buff as contrasted with rich ochraceous; underparts more buffy. Skull: Smaller in every measurement taken; zygomatic arches markedly less widely spreading; braincase narrower and more vaulted; tympanic bullae with a median ventral ridge as opposed to smooth; pterygoid hamulae slenderer; interpterygoid space narrowly V-shaped as opposed to U-shaped; upper incisors smaller and lighter in color.

Compared with topotypes of *Thomomys bottae absonus*, *dissimilis* differs in the following features: Size smaller in every measurement taken. Color: Lighter throughout. Skull: Smaller in every measurement taken, except alveolar length of upper molar series which is greater; skull narrower and weaker; zygomatic arches weaker and less widely spreading; tympanic bullae more ridged on ventral surface and shorter (more rounded) in antero-posterior measurement; upper incisors shorter and narrower; molariform teeth larger.

Thomomys bottae dissimilis resembles *T. b. osgoodi* more than any other subspecies but differs in: Size smaller throughout. Color: Slightly darker dorsally. Skull: Smaller in every measurement taken, and slenderer; rostrum relatively longer; zygomatic arches weaker, and less widely spreading, more converging anteriorly; tympanic bullae less rounded, more ridged medioventrally; upper incisors shorter but narrower; molariform teeth smaller.

Remarks.—The Henry Mountains, in eastern Garfield County, are in the Colorado River drainage. The surrounding country is desertlike and cut by gullies and washes with sheer escarpments and precipitous draws. The type locality of *dissimilis* is possibly in an isolated area. Only three specimens were available to Goldman when he named *dissimilis*. He commented on the close resemblance to *osgoodi* which inhabits the country to the north. I have examined only one of the three specimens available to Goldman. Although I can see the characters that he mentioned, I am not fully convinced that *dissimilis* is separable from *osgoodi*. Two specimens from Escalante, Garfield County, are referred to *absonus*, but they show intergradation with *dissimilis*.

Specimens examined.—One (U. S. N. M.) from E slope Mount Ellen, Henry Mountains, 8,000 ft., Garfield County.

Thomomys bottae aureus Allen

Thomomys aureus Allen, Bull. Amer. Mus. Nat. Hist., 5:49, April 28, 1893.

Thomomys bottae aureus Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935; Benson, Univ. California Publ. Zoöl., 40:450, December 31, 1935.

Thomomys fulvus aureus Goldman, Journ. Washington Acad. Sci., 21:417, October 19, 1931; Journ. Washington Acad. Sci., 23:464, October 15, 1933.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:74, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—No. 5243/4123. American Museum of Natural History; Bluff City, San Juan County, Utah; May 12, 1892; collected by Charles P. Rowley (after Allen, type not seen).

Range.—All of San Juan County (except extreme southwestern part) and Grand County east of the Colorado River.

Diagnosis.—Size large (see measurements). Color: Upper parts Cinnamon Buff, lighter on sides; underparts generally white, or if colored at all with only a faint wash of Light Buff; nose and chin blackish gray; top of head blackish due to admixture of black hairs; postauricular patches small and dusky; front feet and hind feet white. Skull: Long, narrow but massive; zygomatic arches not widely spreading, but heavy; jugals thick, union of jugals and zygomatic processes of maxillae thickened; rostrum long but wide; top of rostrum convex in lateral view; ascending processes of premaxillae wide and heavy; nasals thin proximally; braincase long and narrow; tympanic bullae well inflated ventrally; alveolar length of upper molar series long; molars large; pterygoid hamulae heavy; interpterygoid space U-shaped; palate arched; upper incisors long and wide.

Comparisons.—Compared with topotypes of *Thomomys bottae osgoodi*, *aureus* differs as follows: Size larger in every measurement taken, except tail which is shorter. Color: Darker throughout except on ventral surface which is lighter. Skull: Larger, longer and wider; nasals longer; rostrum wider and longer; zygomatic arches more nearly straight and heavier; ascending processes of premaxillae wider; basioccipital longer; interpterygoid space U-shaped as opposed to V-shaped; tympanic bullae larger; upper incisors longer, wider; molars larger.

Topotypical specimens of *aureus* can be distinguished from those of *Thomomys bottae dissimilis* by: Size larger throughout. Color: A trifle darker on dorsal surface. Skull: Larger in every measurement taken; zygomatic arches heavier and more nearly straight; tympanic bullae larger and more inflated ventrally; interpterygoid space U-shaped as opposed to V-shaped; alveolar length of upper molar series longer; molars larger; upper incisors longer and wider.

Topotypes of *aureus* differ from those of *Thomomys bottae absonus* as follows: Size larger in every measurement taken. Color: Darker dorsally, Light Ochraceous as opposed to Cinnamon Buff; due to admixture of gray, *absonus* has more of a grayish cast. Skull: Larger in every measurement taken, longer, narrower and more compact; zygomatic arches heavier; ascending processes of premaxillae wider; jugals heavier; tympanic bullae larger; interpterygoid space U-shaped rather than V-shaped; upper incisors longer and wider; molars larger.

From topotypes of *Thomomys bottae planirostris*, *aureus* can be distinguished as follows: Size larger; tail shorter. Color: Lighter throughout. Skull: Larger in every measurement taken except zygomatic breadth, extension of premaxillae posterior to nasals, and length of upper molariform series which are less; rostrum longer, wider and more convex; nasals slightly arched rather than straight; depression absent rather than present in posterior region of nasals; zygomatic arches not so widely spreading, but equally heavy.

For comparisons with *Thomomys bottae alexandrae*, see accounts under that form.

Remarks.—Topotypes of *aureus* are among the largest pocket gophers in the state. They are exceeded in total length only by *T. b. lenis* and are approached by *T. b. aureiventris* and *T. b. planirostris*. On the average they have the longest hind foot, body and ear. The length of the skull is second only to that of *lenis* as also is the length and breadth of the rostrum relative to the basilar length.

From the time of the original description of *aureus* in 1893 until 1930, all light colored gophers from Utah were referred to that form. Barnes (1927:100) gives the range of *aureus* as extending completely across southern Utah and on the west and east sides as far north as central Utah. Since 1930, forms named by E. R. Hall, W. H. Burt, E. A. Goldman and the writer have restricted the range of *aureus* in Utah to that part of the state east of the Colorado River.

Specimens examined.—Total, 22, as follows: *San Juan County*: Bluff, 3,300 ft., 22 (15, M. V. Z.).

***Thomomys bottae birdseyei* Goldman**

Thomomys bottae birdseyei Goldman. Proc. Biol. Soc. Washington, 50:134, September 10, 1937.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—Male, adult skin and skull, No. 161654. U. S. National Museum (Biological Surveys Collection); Pine Valley Mountains, 5 mi. E Pine Valley, 8,300 ft., Washington County, Utah; April 10, 1909; collected by Clarence Birdseye; original number 861 (after Goldman, type not seen).

Range.—High mountains and plateaus of southwestern Utah.

Diagnosis.—Size medium (see measurements). Color: Upper parts between Cinnamon and Sayal Brown, finely mixed with black in median dorsal region, grading over sides and flanks to Cinnamon on underparts; front feet, hind feet, and distal part of tail white; postauricular patches, chin, cheeks and top of head grayish black. Skull: Depressed along median line of frontals and posterior ends of nasals; region of nasofrontal suture concave ventrally; zygomatic arches heavy and widely spreading, widest posteriorly; posterior ends of nasals straight, tending to be somewhat rounded in some specimens; extension of premaxillae posterior to nasals moderate; tympanic bullae moderately inflated ventrally; basioccipital wide; interpterygoid space widely V-shaped.

Comparisons.—Topotypes of *birdseyei* differ from near topotypes of *Thomomys bottae virgineus*, from Beaverdam Wash as follows: Size larger; tail and hind foot longer. Color: Darker throughout, between Cinnamon and Sayal Brown as opposed to Cinnamon Buff. Skull: Larger in every measurement taken except extension of premaxillae posterior to nasals, and length and width of rostrum which are less; skull more depressed in nasofrontal region; zygomatic arches more widely spreading; zygomatic processes of squamosals shorter; pterygoid hamulae longer; tympanic bullae smaller and less inflated ventrally.

Among named races of *T. bottae*, *birdseyei* most closely resembles *trumbullensis* in size, but differs as follows: Hind foot and tail longer. Color: Lighter throughout; postauricular patches smaller and lighter. Skull: Larger; mastoid breadth less; zygomatic arches wider and more widely spreading posteriorly; median frontal depression more marked; extension of premaxillae posterior to nasals greater; tympanic bullae less inflated ventrally; molariform teeth larger.

For comparisons with *Thomomys bottae planirostris* see account of that form.

Remarks.—*T. b. birdseyei* is apparently endemic to the mountainous area of southwestern Utah in Washington and Iron counties. It intergrades with *virgineus* and with *planirostris* as described in the account of the latter.

Specimens examined.—Total, 8, distributed as follows: *Washington County*: Pine Valley, 1 (U. S. N. M.); Pine Valley Mountains, 5 mi. E Pine Valley, 8,300 ft., 3 (U. S. N. M.); Pine Valley campground, 6,800 ft., 1 (R. H.); 3/4 mi. E town of Pine Valley, 6,500 ft., 3 (R. H.).

Additional records.—*Washington County*: Hebron, 1; Mountain Meadows, 2 (Bailey 1915:75).

***Thomomys bottae virgineus* Goldman**

Thomomys bottae virgineus Goldman, Proc. Biol. Soc. Washington, 50:133, September 10, 1937.

Type.—Male, adult, skin and skull, No. 262016, U. S. National Museum (Biological Surveys Collection); Beaverdam Creek, near confluence with Virgin River, Littlefield, 1,500 ft., Mohave County, Arizona; October 16, 1936; collected by Luther C. Goldman; original number 67 (after Goldman, type not seen).

Range.—Extreme southwestern Utah, in Beaverdam Wash, Washington County, Utah.

Diagnosis.—Size medium (see measurements). Color: Upper parts Cinnamon Buff, finely mixed with black; sides and flanks Pinkish Buff; underparts Pale Pinkish Buff; front feet, hind feet, and distal part of tail white; nose, cheeks, chin and top of head grayish black. Skull: Robust, with moderately wide zygomatic arches; zygomatic processes of maxillae wide; zygomatic processes of squamosals long; jugals concave laterally, giving the zygomatic arches the appearance of double bowing; nasals long; extension of premaxillae posterior to nasals long; tympanic bullae well inflated ventrally; pterygoid hamulae heavy; interpterygoid space widely V-shaped; molariform teeth large.

Comparisons.—For comparisons of *virgineus* with *Thomomys bottae planirostris* and *T. b. birdseyei* see accounts under those forms.

Topotypical specimens of *virgineus* can be distinguished from those of *Thomomys bottae trumbullensis* as follows: Size smaller. Color: Lighter throughout. Skull: Zygomatic arches less widely spreading; jugals more bowed medially; zygomatic processes of squamosals longer; extension of premaxillae posterior to nasals greater; tympanic bullae larger and more inflated ventrally; molariform teeth larger.

Compared with topotypes of *Thomomys bottae centralis*, *virgineus* differs in: Size smaller; tail shorter; hind foot smaller. Color: Deeper Cinnamon Buff, thus darker in overall appearance. Skull: Smaller, but relatively wider;

zygomatic processes of maxillae heavier; region of maxillo-jugal sutures thicker; jugals more concave laterally; tympanic bullae more inflated ventrally; molariform teeth larger.

Remarks.—This pocket gopher occupies practically the same range in Utah as the large kangaroo rat *Dipodomys deserti deserti* Stephens. Both are found in the Beaverdam Wash. The type locality of *virgineus* is but a short distance down the Beaverdam Creek at Littlefield, Arizona. It intergrades with *birdseyei*, the mountain form to the north and east (see remarks under *birdseyei*). There are evidences of intergradation with *planirostris* of the Virgin River Valley above the narrows of the Virgin River where it cuts through the Beaverdam Mountains (see the discussion under *planirostris*). There are intergradational tendencies exhibited towards *centralis* in some specimens. Some of the animals are practically indistinguishable in color and there are intergrading cranial characters in the nasals, zygomatic arches and tympanic bullae.

Specimens examined.—Total, 20, distributed as follows: *Washington County*: Beaverdam Wash, 8 mi. N Utah-Arizona border, 7; Beaverdam Wash, 5 mi. N Utah-Arizona border, 2,600 ft., 13.

***Thomomys bottae planirostris* Burt**

Thomomys perpallidus planirostris Burt, Proc. Biol. Soc. Washington, 44:38, May 8, 1931.

Thomomys bottae planirostris Hall and Davis, Proc. Biol. Soc. Washington, 47:52, February 9, 1934; Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935; Presnall, Zion-Bryce Mus. Bull., 2:14, January, 1938; Long, Journ. Mamm., 21:176, May 14, 1940.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927; Woodbury, Ecological Monographs, 3:193, April, 1933.

Thomomys bottae centralis Hall and Davis, Proc. Biol. Soc. Washington, 47:52, February 9, 1934; Presnall, Zion-Bryce Mus. Bull., 2:14, January, 1938.

Thomomys perpallidus centralis Hall, Univ. California Publ. Zoöl., 23:445, July 8, 1930.

Thomomys bottae nicholi Goldman, Journ. Washington Acad. Sci., 28:337, July 15, 1938, type from Shivwits Plateau, 20 mi. S Wolf Hole (road to Parashonts), 5,000 ft., Mohave County, Arizona; Hardy, Ecological Monographs, 15:98, January, 1945.

Thomomys bottae trumbullensis Hall and Davis, Proc. Biol. Soc. Washington, 47:52, February 9, 1934.

Type.—Male, adult, skin and skull, No. 8395, Collection of Donald R. Dickey; Zion National Park, Washington County, Utah; May 4, 1920; collected by A. Brazier Howell; original number 2184 (after Burt, type not seen).

Range.—Valley of the Virgin River from Zion National Park west to the Beaverdam Mountains.

Diagnosis.—Size large (see measurements); tail long. Color: Upper parts Sayal Brown; underparts between Vinaceous Cinnamon and Cinnamon, grading to Pinkish Cinnamon in some specimens; nose, chin, cheeks, postauricular patches, and top of head grayish black; front feet and hind feet white; tail Pinkish Buff, with distal third white. Skull: Massive and ridged; nasals straight and flat, simple distally; dorsal surface of rostrum slightly concave at proximal end of nasals; zygomatic arches widely spreading, widest posteriorly; zygomatic processes of maxillae heavy; premaxillae broad and extending far beyond posterior end of nasals; rostrum wide and heavy; palate slightly arched; pterygoid hamulae heavy; interpterygoid space V-shaped; tympanic bullae moderately inflated ventrally, somewhat compressed laterally; upper incisors long and heavy; molariform teeth large.

Comparisons.—Compared with topotypes of *Thomomys bottae birdseyei*, *planirostris* differs as follows: Size larger, except total length which averages slightly less in females. Color: Lighter throughout. Skull: Larger in every measurement taken; more massive; rostrum wider, longer and more nearly flat; nasals straight and not inflated dorsally on distal end; premaxillae wider at posterior ends; extension of premaxillae posterior to nasals greater; zygomatic arches heavier, especially the zygomatic processes of the maxillae; posterior ends of nasals more nearly truncate as opposed to generally rounded; tympanic bullae more nearly flat and relatively smaller; upper incisors longer and heavier; interpterygoid space more narrowly V-shaped; molariform teeth much heavier.

Topotypes of *planirostris* differ from near topotypes of *Thomomys bottae virgineus* as follows: Size larger; tail and hind foot longer. Color: Slightly darker dorsally, but markedly darker ventrally; postauricular patches smaller and lighter. Skull: Larger in every measurement taken; skull more massive; nasals flat, neither arched nor swollen distally; rostrum wider; nasofrontal region flattened or concave as opposed to convex; premaxillae relatively narrower; zygomatic arches heavier, especially in the processes of the maxillae; tympanic bullae smaller and less inflated ventrally; interpterygoid space generally more narrowly V-shaped; upper incisors longer and heavier; molariform teeth larger.

From topotypes of *Thomomys bottae trumbullensis*, *planirostris* differs in: Size larger throughout; tail longer. Color: Much lighter throughout. Skull: More convex dorsally; rostrum wider and more depressed distally; extension of premaxillae posterior to nasals greater; zygomatic arches shorter, and not as widely spreading posteriorly; interpterygoid space more narrowly V-shaped; tympanic bullae smaller; upper incisors wider and longer; molariform teeth larger.

Topotypes of *planirostris* can be easily distinguished from those of *Thomomys bottae absonus* by darker color throughout and markedly larger size.

Remarks.—From the synonymy at the beginning of this account one may note that the animals here ascribed to this subspecies have had nearly as many subspecific names applied to them as there have been investigators who have written about them. Although each of the previous writers had but a small amount of material upon which to base his opinion, the diversity of opinion as to subspecific status bespeaks the instability of these animals. The present study is based upon eighty animals including additional comparative material.

All animals from Zion National Park have the characters pointed out by Burt (1931:38) in his description of this form. Farther down the Virgin River Valley towards St. George, however, some very perplexing problems of intergradation are encountered. St. George and environs may correctly be thought of as a "melting pot." Each of the fifty-seven animals studied from this region is an intergrade; some specimens combine the characters of three subspecies.

As may be seen on the distribution map, three different subspecies of *Thomomys bottae* occur in Washington County. Down the river, below St. George, the race *virgineus* inhabits the Virgin River Valley below the narrows of the Beaverdam Mountains. Because these narrows are filled with water from wall to wall during periods of high runoff, they form an effective barrier at present to migration of pocket gophers. The mountains to the north of St. George are inhabited by the dark form, *birdseyei*. The type locality of *planirostris* is on the middle reaches of the Virgin River, in Zion National Park. In addition Mount Trumbull to the south, in northern Arizona, is the locality of another subspecies, *trumbullensis*.

Unquestionably the easiest route of migration into the St. George area is down the Virgin River from Zion National Park; no barrier to gophers occurs between the Park and St. George. Although the animals from St. George are all intergrades, the majority of their affinities as would be expected are with *planirostris* from Zion National Park. The river itself is not an impassable barrier for gophers to the north and south of it, since this stream frequently changes its course, and often nearly dries up. The Virgin River Valley in Zion National Park is in the bottom of a relatively deep, narrow canyon which has sheer rock escarpments. The upper reaches of the river are inhabited by pocket gophers of another species, *Thomomys talpoides*.

Two specimens from St. George, north of the Virgin River, were identified as *centralis* by Hall and Davis (1934:52), but were stated to be intergrades between *centralis*, *trumbullensis* and *planirostris*. Goldman (1938:338) referred twelve specimens from St. George to *nicholi*, but stated that they intergraded with *planirostris*. Twenty-six other specimens from three miles southwest of St. George on the west side of Santa Clara Creek, about one-half mile above its confluence with the Virgin River and on its north side, like the topotypes of *planirostris* were taken in May and have complete, fresh summer pelage. With the exception of two specimens which show the ventral color of *virgineus*, these animals are indistinguishable in color from the topotypes of *planirostris*. A study of eleven measurements of the males of this series yield the following data: Like *planirostris* in four measurements, *birdseyei* in one, *virgineus* in one; intergrade between *planirostris* and *birdseyei* in two, *planirostris* and *virgineus* in two and *birdseyei* and *virgineus* in one. Corresponding measurements of the females show the animals to be: Like *planirostris* in four measurements, *birdseyei* in one, *virgineus* in two; intergrade between *planirostris* and *birdseyei* in two, *planirostris* and *virgineus* in one and *birdseyei* and *virgineus* in one. In eight of eleven measurements the males either are like *planirostris* or intergrade towards it, and the females are similarly

allied to *planirostris* in seven out of eleven measurements. In none of the measurements was either sex referable to *trumbullensis*.

Intergradation was noted in still other cranial details. In the heavy, relatively straight zygomatic arches, a majority of the skulls resemble those of *planirostris*, although some show the elongated zygomatic processes of the squamosals that are characteristic of *virgineus*. Some skulls show a tendency toward *birdseyei* in the widely spreading posterior regions of the zygomatic arches. The nasals for the most part are as in *planirostris*. Intergradation between all three subspecies is shown in the extension of the premaxillae posterior to the nasals. Some skulls show the lateral concavity of the jugals which is characteristic of *virgineus*. The tympanic bullae are variable but on the average are intermediate between those of *planirostris* and *birdseyei*, but more as in the latter. The size of the pterygoid hamulae is like that of *planirostris*, but the shape of the interpterygoid space is more like that of *birdseyei*. The size of the molariform teeth is as in *birdseyei*. The incisors are intermediate between those of *planirostris* and *birdseyei*, but more like those of *birdseyei*.

Eighteen specimens from St. George and its environs, on the north side of the Virgin River, agree with the twenty-six specimens just described, except that they show more evidence of intergradation with *birdseyei* in slightly darker color, length of hind foot, length of nasals and alveolar length of the upper molar series.

One specimen from three miles south, two from two miles southwest, another from four miles southeast of St. George, and four immature animals from Short Creek Road south of the town of Virgin, all on the south side of the Virgin River, are darker than topotypes of *planirostris* and show intergradation with *trumbullensis* to the south. In size they are likewise closer to the latter race. They intergrade with *trumbullensis* in the size and shape of the zygomatic arches and tympanic bullae. In the majority of cranial details, however, they are like *planirostris* to which they are here referred.

One specimen, a skin only, from Danish Ranch, 5 miles northwest of Leeds, north of the Virgin River is an intergrade in size and color between *birdseyei* and *planirostris*, but referable to the latter.

Three specimens from the East Entrance, and three from near the east entrance to Zion National Park are much darker than topotypes of *planirostris*. All of these animals are in worn pelage, thus allowing a great amount of the black underfur to show, which gives a markedly darker color. The unworn hair is only slightly darker than that of the topotypes. The cranial details prove these animals to be intergrades between *planirostris* and *trumbullensis*. They resemble *trumbullensis* in size of tympanic bullae, extension of the premaxillae posterior to the nasals and shape of the nasals. The majority of the cranial details are as in *planirostris* to which they are here referred.

When Goldman (1938:337) named *Thomomys bottae nicholi* from northern Arizona he referred twelve specimens from St. George, Washington County, Utah, to his newly named race. He noted that the animals from this region intergrade with *planirostris*. I have had occasion to study one-fourth of the material available to Goldman for his original description of *nicholi*. For his specimens listed as from St. George, the exact locality of capture, which is so essential in this distributional study, was not given. All of the specimens that I have seen from the Biological Surveys Collection are from the south side of the Virgin River, while St. George itself is on the north side. As noted earlier in this account there are differences between the gophers from the two sides of the Virgin River in this area. Those from the north side are intergrades between *birdseyei*, *planirostris* and *virgineus*, while those from the south side are intergrades between *planirostris* and *trumbullensis*.

Goldman (*loc. cit.*) mentioned several times that the skulls of *nicholi* were nearly indistinguishable from, or closely resembled those of, *trumbullensis*. Color was the only truly diagnostic character mentioned by Goldman. My study reveals the same differences and likenesses found by Goldman, but I consider color alone insufficient basis in this instance for establishing a new subspecies, and regard *Thomomys bottae nicholi* as a synonym of the earlier proposed name, *Thomomys bottae trumbullensis*.

The animals from the south side of the Virgin River, labelled as from St. George, Washington County, heretofore referred by Goldman to *nicholi*, are intergrades between *trumbullensis* and *planirostris* and along with other specimens from the same place are referable to the latter race.

Specimens examined.—Total, 68, distributed as follows: *Washington County*: Danish Ranch, 5 mi. NW Leeds, 1; Zion National Park, 2 (M. V. Z.); Grotto Camp, Zion National Park, 4,300 ft., 6 (N. H. M. S. D.); Springdale, 3,400 ft., 4 (K. U.); near Short Creek Road, S town of Virgin, 4 (R. H.); St. George, N Virgin River, 2,950 ft., 21 (4, M. V. Z.; 8, R. H.; 9, N. H. M. S. D.); Santa Clara Creek, 3 mi. SW St. George, 2,800 ft., 26; St. George, S Virgin

River, 5 (2, M. V. Z.; 3, U. S. N. M.); 2 mi. SE St. George, 2,950 ft., 2 (N. H. M. S. D.); 3 mi. S St. George, 1 (C. M.); 4 mi. SE St. George, S Virgin River, 1 (R. H.); 6 mi. S St. George, 2,700 ft., 6 (K. U.). *Kane County*: East Entrance Zion National Park, 5,725 ft., 3 (N. H. M. S. D.); near East Entrance Zion National Park, 5,500 ft., 3 (N. H. M. S. D.).

Additional records.—*Washington County*: Zion National Park, 22; Washington, 7 (Burt, 1931:39); St. George, 5; Santa Clara, 2 (Bailey, 1915:75).

***Thomomys bottae absonus* Goldman**

Thomomys perpallidus absonus Goldman, Journ. Washington Acad. Sci., 21:425, October 19, 1931.

Thomomys bottae absonus Hall and Davis, Proc. Biol. Soc. Washington, 47:52, February 9, 1934; Goldman, Proc. Biol. Soc. Washington, 48:156, October 31, 1935.

Thomomys perpallidus aureus Bailey, N. Amer. Fauna, 39:75, November 15, 1915; Barnes, Bull. Univ. Utah, 12 (No. 15):85, April, 1922; Bull. Univ. Utah, 17 (No. 12):100, June, 1927.

Type.—Male, adult, skin and skull, No. 250016, U. S. National Museum (Biological Surveys Collection); Jacobs Pools, Houserock Valley, 4,000 ft., Coconino County, Arizona; June 7, 1931; collected by E. A. Goldman; original number 23569 (after Goldman, type not seen).

Range.—Southern Utah in Kane and Garfield counties, in the drainages of Kanab Creek, Johnson Creek, Paria River and Escalante River.

Diagnosis.—Size medium (see measurements). Color: Upper parts Ochraceous Buff mixed with dusky; sides and underparts Light Ochraceous Buff; chin, nose, cheeks and top of head grayish black; postauricular patches black mixed with buff; front feet, hind feet, inguinal region and distal third of tail white. Skull: Nasals relatively long; rostrum narrow; ascending processes of premaxillae narrow; extension of premaxillae posterior to nasals short; lambdoidal and sagittal crests poorly developed; zygomatic arches light; jugals nearly straight; palate narrow; molariform teeth small.

Comparisons.—Compared with topotypes of *Thomomys bottae trumbullensis*, *absonus* differs in: Size smaller. Color: Markedly lighter throughout. Skull: Smoother, less angular; zygomatic arches weak as opposed to robust; nasals more convex as viewed laterally; extension of premaxillae posterior to nasals less; ascending processes of premaxillae narrower; palate narrower; palatal pits shallower; rostrum narrower; molariform teeth smaller.

For comparisons of *absonus* with *Thomomys bottae aureus* see account under that form.

Among named races of *Thomomys bottae*, *absonus* most closely resembles *planirostris*, but can be distinguished from the topotypes as follows: Size markedly smaller. Color: Lighter, more buffy throughout. Skull: Smaller, less ridged and more nearly flat; nasals convex as opposed to flat; extension of premaxillae posterior to nasals less; width of ascending processes of premaxillae less; zygomatic arches weaker; palate narrower; alveolar length of upper molar series shorter; tympanic bullae more inflated ventrally; molariform teeth smaller and lighter.

Remarks.—One specimen from Kanab is an intergrade between *trumbullensis* and *absonus*. The majority of its characters are with *absonus* to which it is referred (see Hall and Davis, 1934:52). Two specimens from Escalante are intergrades between *absonus* and *dissimilis*, but are referable to *absonus*.

Specimens examined.—Total, 3, distributed as follows: *Garfield County*: Escalante, 5,258 ft., 2 (B. Y. U.), *Kane County*: Kanab, 4,925 ft., 1 (M. V. Z.).

***Thomomys bottae alexandrae* Goldman**

Thomomys alexandrae Goldman, Journ. Washington Acad. Sci., 23:464, October 15, 1933.

Thomomys bottae alexandrae Benson, Univ. California Publ. Zoöl., 40:449, December 31, 1935.

Type.—Male, adult, skin and skull, No. 250969, U. S. National Museum (Biological Surveys Collection); 5 mi. SE Rainbow Lodge, near Navajo Mountain, Coconino County, Arizona; June 16, 1933; collected by E. A. Goldman;

	Total length	Length of tail	Length of hind foot	Basilar length	Length of nasals	Zygomatic breadth	Mastoid breadth	Interorbital breadth	Alveolar length of upper molar series	Extension of premax post. to nasals	Length of rostrum	Breadth of rostrum
<i>T. b. aureiventris</i> , 4; topotypes (Hall, 1930:446)												
Av.	243	67	32	36.4	14.7	26.5	21.5	6.6	7.9	2.4		...
Min.	232	59	31	35.3	14.0	25.5	20.9	6.1	7.8	1.8		...
Max.	253	72	33	37.1	15.3	27.3	22.3	6.9	8.0	3.4		...
<i>T. b. centralis</i> , 9; topotypes (Hall, 1930:446)												
Av.	237	75	30	36.3	14.6	25.2	20.7	6.6	8.0	3.2		...
Min.	215	61	29	34.5	13.9	24.6	19.7	5.8	7.5	2.2		...
Max.	250	83	32	38.0	15.9	26.1	21.9	7.2	8.7	4.5		...
<i>T. b. albicaudatus</i> , 7; topotypes (Hall, 1930:446)												
Av.	228	65	31	35.4	14.0	26.1	20.5	6.6	8.1	3.2		...
Min.	223	59	29	34.9	13.4	24.9	19.8	6.4	7.8	3.0		...
Max.	235	72	32	36.1	15.1	27.8	21.1	6.9	8.4	3.8		...
<i>T. b. robustus</i> , 9; topotypes												

Av.	222	65	29	34.1	13.6	26.0	20.8	6.4	7.8	2.7	15.7	8.4
Min.	214	59	28	32.6	13.0	25.2	20.0	6.1	7.3	2.0	14.7	8.1
Max.	236	70	31	35.7	14.4	26.7	21.5	6.7	8.2	3.0	17.0	8.8
<i>T. b. stansburyi</i> , 5; topotypes												
Av.	206	60	28	32.3	12.4	22.4	19.1	6.3	7.6	2.8	14.7	7.5
Min.	198	58	26	30.6	12.0	21.5	18.2	6.2	7.0	2.5	14.1	7.1
Max.	215	68	31	33.4	13.0	23.1	20.1	6.5	8.0	3.0	15.4	7.8
<i>T. b. nesophilus</i> , 4; topotypes												
Av.	230	69	32	35.3	14.4	25.5	20.4	6.8	8.4	2.5	17.1	8.2
Min.	220	60	30	33.6	14.1	24.9	19.8	6.5	8.2	2.1	16.4	7.6
Max.	242	75	33	36.5	14.8	26.2	21.1	7.1	8.7	2.9	18.4	8.6
<i>T. b. minimus</i> , 2; topotypes												
Av.	184	60	25	30.7	11.3	21.3	18.7	6.4	7.4	2.5	13.9	7.5
Min.	179	55	24	28.7	10.2	20.2	17.8	6.3	7.3	2.5	12.9	7.0
Max.	189	64	26	32.8	12.5	22.4	19.6	6.4	7.6	2.5	15.0	7.9
<i>T. b. lenis</i> , 2; topotypes												
Av.	251	80	32	39.7	16.0	28.6	22.6	6.8	8.3	3.4	18.4	8.8
Min.	248	74	31	39.4	15.8	28.4	22.4	6.6	8.2	3.0	17.9	8.6
Max.	255	86	32	29.9	16.2	28.7	22.7	6.9	8.5	3.7	18.8	8.9
<i>T. b. contractus</i> , 8; topotypes												
Av.	229	74	31	33.3	12.5	23.7	19.1	6.6	7.6	3.0	15.4	7.3
Min.	209	63	28	30.0	10.9	21.4	17.7	6.3	7.2	2.4	13.5	6.5
Max.	255	85	33	37.4	14.5	26.4	20.9	6.9	8.0	3.5	18.2	8.0

MEASUREMENTS OF ADULT MALES OF THOMOMYS—Continued

[illegible]

Av.	228	66	29	34.7	13.5	25.5	20.7	6.6	7.3	2.3	15.7	8.7
Min.	210	60	26	33.0	13.1	24.6	20.1	6.5	7.0	2.2	14.9	8.5
Max.	250	78	30	37.6	14.6	27.0	21.4	6.8	8.0	2.5	17.1	9.0
<i>T. b. planirostris</i> , 8; topotypes (Burt, 1931:39)												
Av.	238	76	32	35.6	13.8	25.9	20.4	6.6	8.5	3.7		8.8
Min.	222	66	31	33.3	12.5	24.4	19.8	6.2	8.2	3.0		8.3
Max.	261	83	34	38.7	15.3	27.6	21.3	7.2	8.9	4.5		9.4
<i>T. b. birdseyei</i> , 3; topotypes												
Av.	227	64	31	34.9	13.8	26.2	20.9	6.2	8.4	2.6	16.3	8.3
Min.	214	52	30	34.5	13.1	26.0	20.1	6.0	8.1	2.2	16.0	8.2
Max.	243	81	32	35.2	14.1	27.4	21.5	6.5	8.8	2.8	16.9	8.4
<i>T. b. virgineus</i> , 5; Beaverdam Wash, 5 mi. N Utah-Arizona Line												
Av.	226	68	29	34.6	13.5	25.6	20.7	6.3	8.0	3.0	16.5	8.5
Min.	216	62	27	33.5	12.8	25.0	20.0	6.1	7.6	2.4	15.3	8.3
Max.	235	70	30	34.9	14.4	26.0	21.1	6.6	8.4	3.5	17.4	8.7
<i>T. b. aureus</i> , 3; topotypes												
Av.	242	68	34	36.6	14.3	25.3	21.4	6.6	8.3	2.4	17.2	8.7
Min.	233	65	32	35.3	13.8	24.6	20.6	6.4	7.7	2.0	16.7	8.3
Max.	251	70	36	37.8	14.9	25.8	22.0	6.8	8.7	2.5	17.9	9.0

	Total length	Length of tail	Length of hind foot	Basilar length	Length of nasals	Zygomatic breadth	Mastoid breadth	Interorbital breadth	Alveolar length of upper molar series	Extension of premax post. to nasals	Length of rostrum	Breadth of rostrum
	<i>T. b. howelli</i> , 5; 10 mi. N Moab											
Av.	213	67	31	33.1	13.5	23.2	20.1	6.5	8.3	2.5	16.1	8.8
Min.	205	64	30	31.8	12.8	22.8	18.9	6.4	8.0	2.3	15.1	8.1
Max.	225	68	32	35.3	14.3	24.1	20.7	6.8	8.8	2.8	17.5	9.4
	No. 3094 (U. U.) <i>T. b. absonus</i> , 1; topotype											
	220	71	29	32.0	13.9	22.6	19.0	6.4	7.0	1.0	15.1	7.2
	No. 158529 (U. S. N. M.) <i>T. b. osgoodi</i> , 1; topotype											
	225	70	29	33.8	13.3	22.7	19.6	6.6	8.4	3.2	16.5	8.3
	<i>T. b. alexandrae</i> , 1; topotype (Benson, 1935:450)											
	205	59	27	33.9	13.7	24.3	19.7	6.5	8.0	...	15.8	8.1

(In millimeters)

T. b. aureiventris, 2; topotypes (Hall, 1930:446)

Av.	212	62	30	32.4	12.9	22.9	19.4	6.7	7.4	2.8		...
Min.	208	58	29	31.8	12.6	22.5	18.9	6.6	7.0	2.7		...
Max.	215	65	30	33.0	13.1	23.3	19.8	6.8	7.8	3.1		...
<i>T. b. centralis</i> , 17; topotypes (Hall, 1930:446)												
Av.	214	67	29	31.8	12.6	22.1	19.0	6.6	7.6	2.7		...
Min.	195	55	27	30.5	11.9	21.3	18.2	5.9	7.0	2.0		...
Max.	229	75	30	33.0	13.8	23.1	20.1	7.1	7.8	3.4		...
<i>T. b. albicaudatus</i> , 4; topotypes (Hall, 1930:446)												
Av.	211	64	30	32.5	12.9	22.9	18.8	6.6	7.7	2.7		...
Min.	199	55	29	31.7	11.9	21.9	18.2	6.1	7.5	2.6		...
Max.	219	70	32	33.8	13.5	24.0	19.5	6.8	8.0	3.0		...
<i>T. b. robustus</i> , 11; topotypes												
Av.	199	61	27	30.6	11.7	22.6	18.8	6.4	7.6	2.6	13.9	7.4
Min.	191	56	22	29.0	10.6	21.0	18.1	6.2	7.1	2.0	12.0	7.1
Max.	207	66	29	31.6	12.2	23.6	19.8	6.7	8.0	2.9	14.7	7.9
<i>T. b. stansburyi</i> , 5; topotypes												
Av.	202	57	28	31.1	12.1	21.9	18.7	6.5	7.7	2.6	14.5	7.4
Min.	195	56	26	29.9	10.6	21.0	17.8	6.2	7.3	2.3	13.4	6.9
Max.	210	63	30	32.7	12.8	22.4	19.5	6.8	8.0	3.0	15.2	7.7
No. 900 (U. U.) <i>T. b. nesophilus</i> , 1; topotype												
	210	65	31	31.2	12.3	23.2	19.3	6.9	8.2	2.2	15.2	7.3
<i>T. b. minimus</i> , 2; topotypes												
Av.	178	56	25	28.2	10.6	19.7	17.4	6.1	7.0	2.3	13.1	6.7
Min.	175	54	24	28.1	10.4	19.6	17.1	6.1	7.0	2.3	13.0	6.5
Max.	181	58	25	28.2	10.8	19.7	17.7	6.1	7.0	2.3	13.2	6.8
<i>T. b. contractus</i> , 6; topotypes												
Av.	219	68	30	33.1	12.6	23.3	19.5	6.5	7.8	2.6	15.5	7.1
Min.	208	58	29	32.2	12.0	22.2	18.9	6.4	7.6	2.3	14.2	7.0
Max.	225	73	31	34.7	13.3	25.2	20.6	6.7	8.2	3.2	17.0	7.3
<i>T. b. levidensis</i> , 4; topotypes												
Av.	205	69	26	30.5	11.1	21.7	17.5	6.6	7.5	2.9	14.0	7.0
Min.	194	61	26	29.3	10.6	21.5	17.3	6.3	7.2	2.8	13.0	6.9
Max.	223	73	27	30.8	11.5	21.9	17.9	6.9	7.8	3.2	14.7	7.2

MEASUREMENTS OF ADULT FEMALES OF THOMOMYS—Continued

	Total Length length of tail	Length of hind foot	Basilar length	Length of nasals	Zygomatic breadth	Mastoid breadth	Interorbital breadth	Alveolar length of upper molar series	Extension of premax post. to nasals	Length of rostrum	Breadth of rostrum	
<i>T. b. convexus</i> , 11; topotypes												
Av.	197	57	27	29.9	12.5	21.7	19.3	6.6	7.7	2.6	14.7	7.4
Min.	182	43	26	27.9	11.2	21.0	18.8	6.2	7.1	2.1	13.3	7.1
Max.	204	63	28	30.9	13.4	22.3	19.8	7.1	7.9	3.1	15.2	7.7
<i>T. b. tivius</i> , 5; topotypes												
Av.	203	68	27	29.5	11.1	21.1	17.8	6.5	7.2	2.4	13.5	6.8
Min.	192	63	26	28.0	10.5	20.1	17.3	6.3	7.1	2.0	12.7	6.4
Max.	215	74	30	31.3	11.4	22.9	19.0	6.7	7.5	3.0	14.2	7.2
<i>T. b. bonnevillei</i> , 7; topotypes												
Av.	199	57	28	31.7	11.8	22.2	19.3	6.6	7.7	3.2	14.9	7.3
Min.	184	50	24	29.4	10.1	20.3	18.1	6.4	7.1	2.6	13.5	6.9
Max.	216	66	29	34.3	13.6	24.3	20.3	7.0	8.5	4.1	16.6	7.7
<i>T. b. sevieri</i> , 7; topotypes												
Av.	205	62	28	30.2	11.8	21.6	18.0	6.4	7.0	2.7	14.2	7.1
Min.	199	54	28	29.4	11.3	20.6	17.7	6.1	6.6	2.1	13.9	6.6
Max.	212	70	29	30.7	12.6	22.1	18.6	6.8	7.4	3.0	14.7	7.6
<i>T. b. wahwahensis</i> , 8; topotypes												
Av.	185	56	27	28.7	11.3	20.6	17.6	6.3	7.1	2.1	12.6	7.1
Min.	180	50	26	26.3	10.2	19.0	16.5	5.8	6.9	1.1	10.8	6.4
Max.	197	62	29	30.7	12.6	22.0	19.0	6.7	7.8	2.9	14.0	7.6
<i>T. b. planirostris</i> , 8; topotypes (Burt, 1931:39)												
Av.	215	71	31	32.2	12.4	23.2	18.7	6.5	8.1	3.6		7.9
Min.	205	61	30	31.5	11.8	22.3	18.1	6.4	7.5	2.8		7.5
Max.	228	78	33	33.0	12.9	24.1	19.5	6.7	8.6	4.5		8.1
<i>T. b. birdseyei</i> , 3; topotypes												
Av.	220	71	29	31.6	11.8	22.7	18.6	6.1	7.4	2.4	14.7	7.5
Min.	217	68	28	31.4	11.0	22.4	18.3	6.0	7.3	1.6	13.3	7.4
Max.	223	75	30	32.0	12.8	23.0	19.1	6.2	7.4	3.0	15.3	7.5
<i>T. b. virgineus</i> , 4; Beaverdam Wash, 5 mi. N Utah-Arizona Line												
Av.	211	64	29	31.6	12.2	22.6	19.4	5.9	7.5	3.1	15.1	7.3
Min.	202	60	27	31.3	11.3	22.4	18.8	5.8	7.3	2.4	14.4	7.2
Max.	218	68	30	32.1	12.8	22.7	20.0	6.1	7.8	3.7	15.5	7.6
<i>T. b. aureus</i> , 3; topotypes												
Av.	226	57	31	33.2	13.3	23.8	19.8	6.7	8.2	1.9	15.3	8.2
Min.	217	54	30	32.8	12.5	23.3	19.6	6.4	8.0	1.6	14.5	8.2
Max.	233	64	31	34.0	14.2	24.4	19.8	6.9	8.4	2.0	16.4	8.3
No. 20300 (C. M.) <i>T. b. howelli</i> , 1; 10 mi. N Moab												
	202	59	28	32.4	12.3	21.1	19.2	6.4	7.9	2.4	15.8	8.3

MEASUREMENTS OF ADULT FEMALES OF THOMOMYS—Concluded

Total Length length of tail	Length of hind foot	Basilar length	Length of nasals	Zygomatic breadth	Mastoid breadth	Interorbital breadth	Alveolar length of upper	Extension of premax	Length of rostrum	Breadth of rostrum
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								molar series	post. to nasals			
			No. 158524 (U. S. N. M.) <i>T. b. dissimilis</i> , 1; topotype									
	188	61	27	28.2	10.1	19.0	16.7	6.1	7.4	2.1	12.8	6.5
				No. 158528 (U. S. N. M.) <i>T. b. osgoodi</i> , 1; topotype								
	203	61	27	29.6	11.5		18.3	6.9	7.4	2.0	14.0	7.3
				<i>T. b. alexandrae</i> , 3; topotypes								
Av.	205	63	28	30.9	11.8	20.8	17.9	6.4	7.6	1.8	14.1	7.5
Min.	195	57	27	28.7	11.5	20.5	17.2	6.3	7.5	1.5	13.6	7.2
Max.	215	70	29	31.5	12.1	22.2	18.6	6.5	7.7	2.0	14.7	7.7

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Transcriber's Notes

The [Contents](#) are for the entire Vol. 1, not the contents of this individual publication.

Made minor punctuation corrections, and the following changes:

Page [11](#): Changed Oquirrah Mountains to Oquirrh Mountains.

Page [15](#): Changed interptergoid to interpterygoid.

Page [25](#): Changed acccounts to accounts.

Page [30](#): Changed distiguished to distinguished.

Page [54](#): Changed hpyothesis to hypothesis.

Page [57](#): Changed under parts to underparts.

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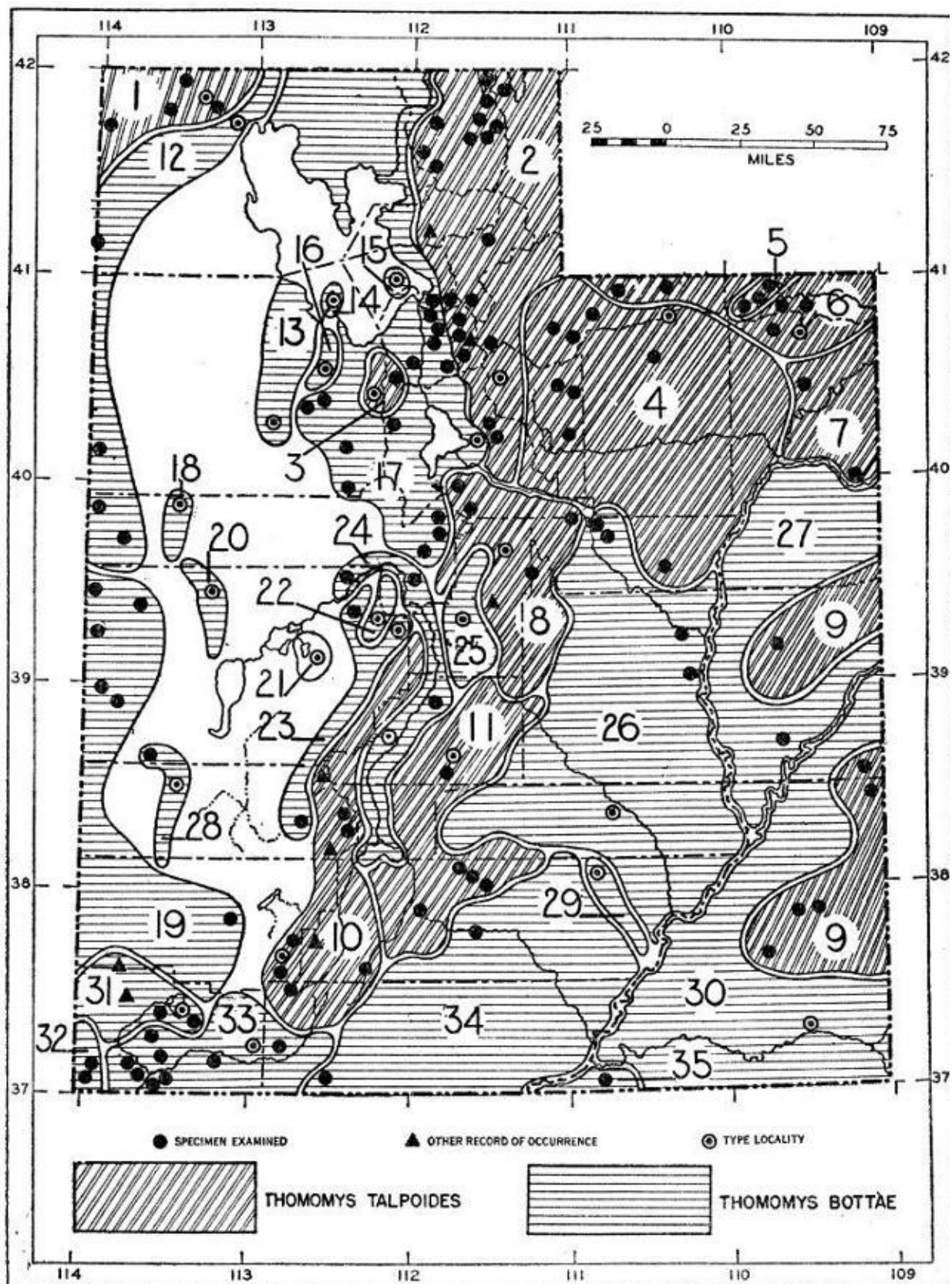
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